

SECTION 5

FLOATING POINT INSTRUCTIONS

This section contains information about the floating-point instructions for the MC68881, MC68882, and MC68040. In this section, all references to the MC68040 do not include the MC68LC040 and MC68EC040. Each instruction is described in detail, and the instruction descriptions are arranged in alphabetical order by instruction mnemonic.

All floating-point instructions apply to the MC68881 and MC68882 processors. The MC68040 directly supports part of the floating-point instructions through hardware. It indirectly supports the remainder by providing special traps and/or stack frames for the unimplemented instructions and data types. The following identification is noted under the instruction title for the MC68040:

Directly Supported—(MC6888X, MC68040)

Software Supported—(MC6888X, MC68040FPSW)

For all MC68040 floating-point instructions, the coprocessor ID field must be 001.

Table 5-1 lists the floating-point instructions directly supported by the MC68040, and Table 5-2 lists the floating-point instructions indirectly supported.

Table 5-1. Directly Supported Floating-Point Instructions

Mnemonic	Description
FABS	Floating-Point Absolute Value
FADD	Floating-Point Add
FBcc	Floating-Point Branch Conditionally
FCMP	Floating-Point Compare
FDBcc	Floating-Point Test Condition, Decrement, and Branch
FDIV	Floating-Point Divide
FMOVE	Move Floating-Point Data Register
FMOVE	Move Floating-Point System Control Register
FMOVEM	Move Multiple Floating-Point System Data Register
FMOVEM	Move Multiple Floating-Point Control Data Register
FMUL	Floating-Point Multiply
FNEG	Floating-Point Negate
FNOP	No Operation
FRESTORE*	Restore Internal Floating-Point State*
FSAVE*	Save Internal Floating-Point State*
FScC	Set According to Floating-Point Condition
FSORT	Floating-Point Square Root
FSUB	Floating-Point Subtract
FSGLDIV	Floating-Point Single-Precision Divide
FSFLMUL	Floating-Point Single-Precision Multiply
FTRAPcc	Trap on Floating-Point Condition
FTST	Test Floating-Point Operand

*These are privileged instructions; refer to **Section 6 Supervisor (Privileged) Instructions** for detailed information.

Table 5-2. Indirectly Supported Floating-Point Instructions

Mnemonic	Description
FACOS	Floating-Point Arc Cosine
FASIN	Floating-Point Arc Sine
FATAN	Floating-Point Arc Tangent
FATANH	Floating-Point Hyperbolic Arc Tangent
FCOS	Floating-Point Cosine
FCOSH	Floating-Point Hyperbolic Cosine
FETOX	Floating-Point e^x
FETOXM1	Floating-Point $e^x - 1$
FGETEXP	Floating-Point Get Exponent
FGETMAN	Floating-Point Get Mantissa
FINT	Floating-Point Integer Part
FINTRZ	Floating-Point Integer Part, Round-to- Zero
FLOG10	Floating-Point Log10
FLOG2	Floating-Point Log2
FLOGN	Floating-Point Loge
FLOGNP1	Floating-Point $\text{Log}_e (x + 1)$
FMOD	Floating-Point Modulo Remainder
FMOVECR	Floating-Point Move Constant ROM
FREM	Floating-Point IEEE Remainder
FSCALE	Floating-Point Scale Exponent
FSIN	Floating-Point Sine
FSINCOS	Floating-Point Simultaneous Sine and Cosine
FSINH	Floating-Point Hyperbolic Sine
FTAN	Floating-Point Tangent
FTANH	Floating-Point Hyperbolic Tangent
FTENTOX	Floating-Point 10^x
FTWOTOX	Floating-Point 2^x



FABS

Floating-Point Absolute Value (MC6888X, MC68040)

FABS

Operation: Absolute Value of Source → FPn

Assembler

Syntax: FABS. < fmt > < ea > ,FPn
FABS.X FPm,FPn
FABS.X FPn
*FrABS. < fmt > < ea > ,FPn
*FrABS.X FPm,FPn
*FrABS.X Pn
where r is rounding precision, S or D
*Supported by MC68040 only.

Attributes: Format = (Byte, Word, Long, Single, Quad, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and stores the absolute value of that number in the destination floating-point data register.

FABS will round the result to the precision selected in the floating-point control register. FSABS and FDABS will round the result to single or double precision, respectively, regardless of the rounding precision selected in the floating-point control register.

Operation Table:

DESTINATION	SOURCE								
	+	In Range	-	+	Zero	-	+	Infinity	-
Result	Absolute Value		Absolute Value		Absolute Value				

NOTE: If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information

FABS

Floating-Point Absolute Value
(MC6888X, MC68040)

FABS

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers
	OPERR	Cleared
	OVFL	Cleared
	UNFL	If the source is an extended-precision denormalized number, refer to exception processing in the appropriate user's manual; cleared otherwise.
	DZ	Cleared
	INEX2	Cleared
	INEX1	If < fmt > is packed, refer to exception processing in the appropriate user's manual; cleared otherwise.
Accrued Exception Byte:	Affected as described in exception processing; refer to the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS MODE REGISTER					
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

FABS

Floating-Point Absolute Value (MC6888X, MC68040)

FABS

Instruction Fields:

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field specifies the location of the source operand. Only data addressing modes can be used as listed in the following table:

Addressing Mode	Mode	Register
Dn	000	reg. number:Dn
An	—	—
(An)	010	reg. number:An
(An) +	011	reg. number:An
– (An)	100	reg. number:An
(d ₁₆ ,An)	101	reg. number:An
(d ₈ ,An,Xn)	110	reg. number:An
(bd,An,Xn)	110	reg. number:An
([bd,An,Xn],od)	110	reg. number:An
([bd,An],Xn,od)	110	reg. number:An

Addressing Mode	Mode	Register
(xxx).W	111	000
(xxx).L	111	001
# < data >	111	100
(d ₁₆ ,PC)	111	010
(d ₈ ,PC,Xn)	111	011
(bd,PC,Xn)	111	011
([bd,PC,Xn],od)	111	011
([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

FABS

Floating-Point Absolute Value
(MC6888X, MC68040)

FABS

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)*
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

*This encoding will cause an unimplemented data type exception in the MC68040 to allow emulation in software.

Destination Register field—Specifies the destination floating- point data register.

Opmode field—Specifies the instruction and rounding precision.

0011000	FABS	Rounding precision specified by the floating-point control register.
1011000	FSABS	Single-precision rounding specified.
1011100	FDABS	Double-precision rounding specified.

FACOS

Arc Cosine
(MC6888X, M68040FPSP)

FACOS

Operation: Arc Cosine of Source $\rightarrow$ FPn

Assembler FACOS. < fmt > < ea > ,FPn

Syntax: FACOS.X FPM,FPn
FACOS.X FPN

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates the arc cosine of that number. Stores the result in the destination floating-point data register. This function is not defined for source operands outside of the range $[-1 \dots +1]$; if the source is not in the correct range, a NAN is returned as the result and the OPERR bit is set in the floating-point status register. If the source is in the correct range, the result is in the range of $[0 \dots \pi]$.

Operation Table:

DESTINATION	SOURCE								
	+	In Range	–	+	Zero	–	+	Infinity	–
Result		Arc Cosine			$+ \pi/2$			NAN	

NOTES:

1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
2. Sets the OPERR bit in the floating-point status register exception byte.

Floating-Point Status Register:

Condition Codes: Affected as described in **3.6.2 Conditional Testing**.

Quotient Byte: Not affected.

Exception Byte:

BSUN	Cleared
SNAN	Refer to 1.6.5 Not-A-Numbers .
OPERR	Set if the source is infinity, $> +1$ or < -1 ; cleared otherwise.
OVFL	Cleared
UNFL	Cleared
DZ	Cleared
INEX2	Refer to inexact result in the appropriate user's manual.
INEX1	If < fmt > is packed, refer to inexact result on decimal input in the appropriate user's manual; cleared otherwise.

Accrued Exception Byte: Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.

FACOS

Arc Cosine (MC6888X, M68040FPSP)

FACOS

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	1	1	0	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn
An	—	—
(An)	010	reg. number:An
(An) +	011	reg. number:An
– (An)	100	reg. number:An
(d ₁₆ ,An)	101	reg. number:An
(d ₈ ,An,Xn)	110	reg. number:An
(bd,An,Xn)	110	reg. number:An
([bd,An,Xn],od)	110	reg. number:An
([bd,An],Xn,od)	110	reg. number:An

Addressing Mode	Mode	Register
(xxx).W	111	000
(xxx).L	111	001
# < data >	111	100
(d ₁₆ ,PC)	111	010
(d ₈ ,PC,Xn)	111	011
(bd,PC,Xn)	111	011
([bd,PC,Xn],od)	111	011
([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

FACOS

Arc Cosine (MC6888X, M68040FPSP)

FACOS

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating-point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is then written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FADD

Floating-Point Add (MC6888X, MC68040)

FADD

Operation: Source + FPn → FPn

Assembler Syntax: FADD. < fmt > < ea > ,FPn
 FADD.X FPm,FPn
 *FrADD. < fmt > < ea > ,FPn
 *FrADD.X FPm,FPn
 where r is rounding precision, S or D
 *Supported by MC68040 only.

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and adds that number to the number contained in the destination floating-point data register. Stores the result in the destination floating-point data register.

FADD will round the result to the precision selected in the floating-point control register. FSADD and FDADD will round the result to single or double-precision, respectively, regardless of the rounding precision selected in the floating-point control register.

Operation Table:

DESTINATION	SOURCE ¹					
	+	In Range	–	+	Zero	–
In Range	+	Add		+	Add	
	–			–		
Zero	+	Add		+	+ 0.0	0.0 ²
	–			–	0.0 ²	– 0.0
Infinity	+	+ inf		+	+ inf	NAN ³
	–	– inf		–	– inf	NAN [‡]

1. If either operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
 2. Returns + 0.0 in rounding modes RN, RZ, and RP; returns – 0.0 in RM.
 3. Sets the OPERR bit in the floating-point status register exception byte.

FADD

Floating-Point Add (MC6888X, MC68040)

FADD

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the source and the destination are opposite-signed infinities; cleared otherwise.
	OVFL	Refer to exception processing in the appropriate user's manual.
	UNFL	Refer to exception processing in the appropriate user's manual.
	DZ	Cleared
	INEX2	Refer to exception processing in the appropriate user's manual.
	INEX1	If < fmt > is packed, refer to exception processing in the appropriate user's manual; cleared otherwise.
Accrued Exception Byte:	Affected as described in exception processing in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS MODE REGISTER					
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

Instruction Fields:

Effective Address field—Determines the addressing mode for external operands.
If R/M = 0, this field is unused and should be all zeros.

FADD

Floating-Point Add (MC6888X, MC68040)

FADD

If R/M = 1, specifies the location of the source operand location. Only data addressing modes can be used as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

- 0 — The operation is register to register.
- 1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)*
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

*This encoding will cause an unimplemented data type exception to allow emulation in software.

Destination Register field—Specifies the destination floating- point data register.

Opmode field—Specifies the instruction and rounding precision.

- 0100010 FADD Rounding precision specified by the floating-point control register.
- 1100010 FSADD Single-precision rounding specified.
- 1100110 FDADD Double-precision rounding specified.

FASIN

Arc Sine (MC6888X, M68040FPSP)

FASIN

Operation: Arc Sine of the Source $\rightarrow$ FPN

Assembler Syntax: FASIN. < fmt > < ea > ,FPn
FASIN.X FPM,FPn
FASIN.X FPN

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates the arc sine of the number. Stores the result in the destination floating-point data register. This function is not defined for source operands outside of the range $[-1 \dots +1]$; if the source is not in the correct range, a NAN is returned as the result and the OPERR bit is set in the floating-point status register. If the source is in the correct range, the result is in the range of $[-\pi/2 \dots +\pi/2]$.

Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	-	+	Zero	-	+	Infinity	-
Result	Arc Sine		+ 0.0		- 0.0		NAN ²		

- NOTES:
- 1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
 - 2. Sets the OPERR bit in the floating-point status register exception byte.

FASIN

Arc Sine (MC6888X, M68040FPSP)

FASIN

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the source is infinity, $> + 1$ or $< - 1$; cleared otherwise
	OVFL	Cleared
	UNFL	Can be set for an underflow condition.
	DZ	Cleared
	INEX2	Refer to inexact result in the appropriate user's manual.
	INEX1	If <code>< fmt ></code> is packed, refer to inexact result on decimal input in the appropriate user's manual; cleared otherwise.
Accrued Exception Byte:	Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	1	0	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

FASIN

Arc Sine (MC6888X, M68040FPSP)

FASIN

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

- 0 — The operation is register to register.
- 1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is then written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FATAN

Arc Tangent (MC6888X, M68040FPSP)

FATAN

Operation: Arc Tangent of Source → FPn

Assembler Syntax: FATAN. < fmt > < ea > ,FPn
 FATAN.X FPm,FPn
 FATAN.X FPm,FPnz

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates the arc tangent of that number. Stores the result in the destination floating-point data register. The result is in the range of $[-\pi/2 \dots +\pi/2]$.

Operation Table:

DESTINATION	SOURCE								
	+	In Range	–	+	Zero	–	+	Infinity	–
Result	Arc Tangent		+ 0.0		– 0.0		+ $\pi/2$		– $\pi/2$

NOTE: If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.

Floating-Point Status Register:

Condition Codes: Affected as described in **3.6.2 Conditional Testing**.

Quotient Byte: Not affected.

Exception Byte:

BSUN	Cleared
SNAN	Refer to 1.6.5 Not-A-Numbers .
OPERR	Cleared
OVFL	Cleared
UNFL	Refer to underflow in the appropriate user’s manual.
DZ	Cleared
INEX2	Refer to inexact result in the appropriate user’s manual.
INEX1	If < fmt > is packed, refer to inexact result on decimal input in the appropriate user’s manual; cleared otherwise.

Accrued Exception Byte: Affected as described in IEEE exception and trap compatibility in the appropriate user’s manual.

FATAN

Arc Tangent
(MC6888X, M68040FPSP)

FATAN

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	1	0	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn
An	—	—
(An)	010	reg. number:An
(An) +	011	reg. number:An
– (An)	100	reg. number:An
(d ₁₆ ,An)	101	reg. number:An
(d ₈ ,An,Xn)	110	reg. number:An
(bd,An,Xn)	110	reg. number:An
([bd,An,Xn],od)	110	reg. number:An
([bd,An],Xn,od)	110	reg. number:An

Addressing Mode	Mode	Register
(xxx).W	111	000
(xxx).L	111	001
# < data >	111	100
(d ₁₆ ,PC)	111	010
(d ₈ ,PC,Xn)	111	011
(bd,PC,Xn)	111	011
([bd,PC,Xn],od)	111	011
([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

FATAN

Arc Tangent (MC6888X, M68040FPSP)

FATAN

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating-point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is then written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FATANH

Hyperbolic Arc Tangent (MC6888X, M68040FPSP)

FATANH

Operation: Hyperbolic Arc Tangent of Source → FPn

Assembler FATANH. < fmt > < ea > ,FPn

Syntax: FATANH.X FPm,FPn
FATANH.X FPn

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates the hyperbolic arc tangent of that value. Stores the result in the destination floating-point data register. This function is not defined for source operands outside of the range (− 1... + 1); and the result is equal to − infinity or + infinity if the source is equal to + 1 or − 1, respectively. If the source is outside of the range [− 1... + 1], a NAN is returned as the result, and the OPERR bit is set in the floating-point status register.

Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	−	+	Zero	−	+	Infinity	−
Result		Hyperbolic Arc Tangent		+	0.0		−	0.0	NAN ²

- NOTES:
- 1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
 - 2. Sets the OPERR bit in the floating-point status register exception byte.

Floating-Point Status Register:

Condition Codes: Affected as described in **3.6.2 Conditional Testing**.

Quotient Byte: Not affected.

FATANH

Hyperbolic Arc Tangent (MC6888X, M68040FPSP)

FATANH

Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the source is $> + 1$ or $< - 1$; cleared otherwise.
	OVFL	Cleared
	UNFL	Refer to underflow in the appropriate user's manual.
	DZ	Set if the source is equal to $+ 1$ or $- 1$; cleared otherwise.
	INEX2	Refer to inexact result in the appropriate user's manual.
	INEX1	If $< \text{fmt} >$ is packed, refer to inexact result on decimal input in the appropriate user's manual; cleared otherwise.

Accrued Exception Byte: Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
											MODE		REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	1	0	1

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

FATANH

Hyperbolic Arc Tangent (MC6888X, M68040FPSP)

FATANH

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

- 0 — The operation is register to register.
- 1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is then written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FBcc

Floating-Point Branch Conditionally

(MC6888X, MC68040)

FBcc

Operation: If Condition True
 Then $PC + d_n \rightarrow PC$

Assembler:
Syntax: FBcc. < size > , < label >

Attributes: Size = (Word, Long)

Description: If the specified floating-point condition is met, program execution continues at the location (PC) + displacement. The displacement is a twos-complement integer that counts the relative distance in bytes. The value of the program counter used to calculate the destination address is the address of the branch instruction plus two. If the displacement size is word, then a 16- bit displacement is stored in the word immediately following the instruction operation word. If the displacement size is long word, then a 32-bit displacement is stored in the two words immediately following the instruction operation word. The conditional specifier cc selects any one of the 32 floating- point conditional tests as described in **3.6.2 Conditional Testing**.

Floating-Point Status Register:

Condition Codes:	Not affected.	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Set if the NAN condition code is set and the condition selected is an IEEE nonaware test.
	SNAN	Not Affected.
	OPERR	Not Affected.
	OVF	Not Affected.
	UNFL	Not Affected.
	DZ	Not Affected.
	INEX2	Not Affected.
	INEX1	Not Affected.
Accrued Exception Byte:	The IOP bit is set if the BSUN bit is set in the exception byte. No other bit is affected.	



FBcc

Floating-Point Branch Conditionally
(MC6888X, MC68040)

FBcc

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	1	SIZE	CONDITIONAL PREDICATE					
16-BIT DISPLACEMENT OR MOST SIGNIFICANT WORD OF 32-BITDISPLACEMENT															
LEAST SIGNIFICANT WORD OF 32-BIT DISPLACEMENT (IF NEEDED)															

Instruction Fields:

- Size field—Specifies the size of the signed displacement.
If Format = 0, then the displacement is 16 bits and is sign- extended before use.
If Format = 1, then the displacement is 32 bits.
- Conditional Predicate field—Specifies one of 32 conditional tests as defined in **Table 3-23 Floating-Point Conditional Tests**.

NOTE

When a BSUN exception occurs, the main processor takes a preinstruction exception. If the exception handler returns without modifying the image of the program counter on the stack frame (to point to the instruction following the FBcc), then it must clear the cause of the exception (by clearing the NAN bit or disabling the BSUN trap), or the exception will occur again immediately upon return to the routine that caused the exception.

FCMP

Floating-Point Compare

(MC6888X, MC68040)

FCMP

Operation:

FPn – Source

Assembler

FCMP. < fmt > < ea > ,FPn

Syntax:

FCMP.X FPm,FPn

Attributes:

Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and subtracts the operand from the destination floating- point data register. The result of the subtraction is not retained, but it is used to set the floating-point condition codes as described in **3.6.2 Conditional Testing**.

Operation Table: The entries in this operation table differ from those of the tables describing most of the floating-point instructions. For each combination of input operand types, the condition code bits that may be set are indicated. If the name of a condition code bit is given and is not enclosed in brackets, then it is always set. If the name of a condition code bit is enclosed in brackets, then that bit is either set or cleared, as appropriate. If the name of a condition code bit is not given, then that bit is always cleared by the operation. The infinity bit is always cleared by the FCMP instruction since it is not used by any of the conditional predicate equations. Note that the NAN bit is not shown since NANs are always handled in the same manner (as described in **1.6.5 Not-A-Numbers**).

DESTINATION	SOURCE											
	+	In Range		–	+	Zero		–	+	Infinity		–
In Range	+	{NZ}		none	none			none	N			none
	–	N		{NZ}	N			N	N			none
Zero –	+	N		none	Z			Z	N			none
		N		none	NZ			NZ	N			none
Infinity	+	none		none	none			none	Z			none
	–	N		N	N			N	N			NZ

NOTE: If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.



FCMP

Floating-Point Compare (MC6888X, MC68040)

FCMP

Floating-Point Status Register:

- Condition Codes: Affected as described in the preceding operation table.
- Quotient Byte: Not affected.
- Exception Byte:

BSUNCleared

SNANRefer to **1.6.5 Not-A-Numbers**.

OPERRCleared

OVFLCleared

UNFLCleared

DZCleared

INEX2Cleared

INEX1If < fmt > is packed, refer to exception processing in the appropriate user's manual; cleared otherwise.
- Accrued Exception Byte: Affected as described in exception processing in the appropriate user's manual.

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE	REGISTER				
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	1	0	0

FCMP

Floating-Point Compare (MC6888X, MC68040)

FCMP

Instruction Fields:

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, specifies the location of the source operand location. Only data addressing modes can be used as listed in the following table:

Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn
An	—	—
(An)	010	reg. number:An
(An) +	011	reg. number:An
– (An)	100	reg. number:An
(d ₁₆ ,An)	101	reg. number:An
(d ₈ ,An,Xn)	110	reg. number:An
(bd,An,Xn)	110	reg. number:An
([bd,An,Xn],od)	110	reg. number:An
([bd,An],Xn,od)	110	reg. number:An

Addressing Mode	Mode	Register
(xxx).W	111	000
(xxx).L	111	001
# < data >	111	100
(d ₁₆ ,PC)	111	010
(d ₈ ,PC,Xn)	111	011
(bd,PC,Xn)	111	011
([bd,PC,Xn],od)	111	011
([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)*

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

*This encoding in the MC68040 will cause an unimplemented data type exception to allow emulation in software.

Destination Register field—Specifies the destination floating- point data register.

FCOS

Cosine
(MC6888X, M68040FPSP)

FCOS

Operation: Cosine of Source $\rightarrow$ FPn

Assembler FCOS. < fmt > < ea > ,FPn

Syntax: FCOS.X Fpm,FPn FCOS.X FPn

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates the cosine of that number. Stores the result in the destination floating-point data register. This function is not defined for source operands of $\pm$ infinity. If the source operand is not in the range of $[-2\pi \dots +2\pi]$, then the argument is reduced to within that range before the cosine is calculated. However, large arguments may lose accuracy during reduction, and very large arguments (greater than approximately 10^{20}) lose all accuracy. The result is in the range of $[-1 \dots +1]$.

Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	-	+	Zero	-	+	Infinity	-
Result	Cosine			+ 1.0			NAN ²		

NOTES:

1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
2. Sets the OPERR bit in the floating-point status register exception byte.

FCOS

Cosine
(MC6888X, M68040FPSP)

FCOS

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Not affected	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the source operand is $\pm$ infinity; cleared otherwise.
	OVFL	Cleared
	UNFL	Cleared
	DZ	Cleared
	INEX2	Refer to inexact result in the appropriate user's manual.
	INEX1	If < fmt > is packed, refer to inexact result on decimal input in the appropriate user's manual; cleared otherwise.
Accrued Exception Byte:	Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
											MODE		REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	1	1	0	1

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

FCOS

Cosine (MC6888X, M68040FPSP)

FCOS

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should contain zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FCOSH

Hyperbolic Cosine

(MC6888X, M68040FPSP)

FCOSH

Operation:

Hyperbolic Cosine of Source → FPn

Assembler

FCOSH. < fmt > < ea > ,FPn

Syntax:

FCOSH.X FPm,FPn FCOSH.X FPn

Attributes:

Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description:

Converts the source operand to extended precision (if necessary) and calculates the hyperbolic cosine of that number. Stores the result in the destination floating-point data register.

Operation Table:

DESTINATION	SOURCE								
	+	In Range	–	+	Zero	–	+	Infinity	–
Result	Hyperbolic Cosine				+ 1.0			+ inf	

NOTE: If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.

Floating-Point Status Register:

Condition Codes:

Affected as described in **3.6.2 Conditional Testing**.

Quotient Byte:

Not affected.

Exception Byte:

BSUN

Cleared

SNAN

Refer to **1.6.5 Not-A-Numbers**.

OPERR

Cleared

OVFL

Refer to overflow in the appropriate user’s manual.

UNFL

Cleared

DZ

Cleared

INEX2

Refer to inexact result in the appropriate user’s manual.

INEX1

If < fmt > is packed, refer to inexact result on decimal input in the appropriate user’s manual; cleared otherwise.

Accrued Exception Byte:

Affected as described in IEEE exception and trap compatibility in the appropriate user’s manual.



FCOSH

Hyperbolic Cosine (MC6888X, M68040FPSP)

FCOSH

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	1	1	0	1

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn
An	—	—
(An)	010	reg. number:An
(An) +	011	reg. number:An
– (An)	100	reg. number:An
(d ₁₆ ,An)	101	reg. number:An
(d ₈ ,An,Xn)	110	reg. number:An
(bd,An,Xn)	110	reg. number:An
([bd,An,Xn],od)	110	reg. number:An
([bd,An],Xn,od)	110	reg. number:An

Addressing Mode	Mode	Register
(xxx).W	111	000
(xxx).L	111	001
# < data >	111	100
(d ₁₆ ,PC)	111	010
(d ₈ ,PC,Xn)	111	011
(bd,PC,Xn)	111	011
([bd,PC,Xn],od)	111	011
([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

FCOSH

Hyperbolic Cosine (MC6888X, M68040FPSP)

FCOSH

R/M field—Specifies the source operand address mode.

- 0 — The operation is register to register.
- 1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FDBcc

Floating-Point Test Condition, Decrement, and Branch (MC6888X, MC68040)

FDBcc

Operation: If Condition True
Then No Operation
Else $D_n - 1 \rightarrow D_n$
If $D_n \neq -1$
Then $PC + d_n \rightarrow PC$
Else Execute Next Instruction

Assembler

Syntax: FDBcc D_n , < label >

Attributes: Unsized

Description: This instruction is a looping primitive of three parameters: a floating-point condition, a counter (data register), and a 16-bit displacement. The instruction first tests the condition to determine if the termination condition for the loop has been met, and if so, execution continues with the next instruction in the instruction stream. If the termination condition is not true, the low-order 16 bits of the counter register are decremented by one. If the result is -1 , the count is exhausted, and execution continues with the next instruction. If the result is not equal to -1 , execution continues at the location specified by the current value of the program counter plus the sign-extended 16-bit displacement. The value of the program counter used in the branch address calculation is the address of the displacement word.

The conditional specifier cc selects any one of the 32 floating-point conditional tests as described in **3.6.2 Conditional Testing**.

Floating-Point Status Register:

Condition Codes: Not affected.

Quotient Byte: Not affected.

Exception Byte:	BSUN	Set if the NAN condition code is set and the condition selected is an IEEE nonaware test.
	SNAN	Not Affected.
	OPERR	Not Affected.
	OVFL	Not Affected.
	UNFL	Not Affected.
	DZ	Not Affected.
	NEX2	Not Affected.
	INEX1	Not Affected.

Accrued Exception Byte: The IOP bit is set if the BSUN bit is set in the exception byte. No other bit is affected.

FDBcc

Floating-Point Test Condition, Decrement, and Branch (MC6888X, MC68040)

FDBcc

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	1	0	0	1	COUNT REGISTER		
0	0	0	0	0	0	0	0	0	0	CONDITIONAL PREDICATE					
16-BIT DISPLACEMENT															

Instruction Fields:

- Count Register field—Specifies data register that is used as the counter.
- Conditional Predicate field—Specifies one of the 32 floating-point conditional tests as described in **3.6.2 Conditional Testing**.
- Displacement field—Specifies the branch distance (from the address of the instruction plus two) to the destination in bytes.

NOTE

The terminating condition is like that defined by the UNTIL loop constructs of high-level languages. For example: FDBOLT can be stated as "decrement and branch until ordered less than".

There are two basic ways of entering a loop: at the beginning or by branching to the trailing FDBcc instruction. If a loop structure terminated with FDBcc is entered at the beginning, the control counter must be one less than the number of loop executions desired. This count is useful for indexed addressing modes and dynamically specified bit operations. However, when entering a loop by branching directly to the trailing FDBcc instruction, the count should equal the loop execution count. In this case, if the counter is zero when the loop is entered, the FDBcc instruction does not branch, causing a complete bypass of the main loop.

When a BSUN exception occurs, a preinstruction exception is taken by the main processor. If the exception handler returns without modifying the image of the program counter on the stack frame (to point to the instruction following the FDBcc), then it must clear the cause of the exception (by clearing the NAN bit or disabling the BSUN trap), or the exception will occur again immediately upon return to the routine that caused the exception.

FDIV

Floating-Point Divide
(MC6888X, MC68040)

FDIV

Operation:

FPn ÷ Source → FPn

Assembler

FDIV. < fmt > < ea > ,FPn

Syntax:

FDIV.X Fm,FPn

*FrDIV. < fmt > < ea > ,FPn

*FrDIV.X Fm,FPn

where r is rounding precision, S or D

*Supported by MC68040 only

Attributes:

Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and divides that number into the number in the destination floating-point data register. Stores the result in the destination floating-point data register.

FDIV will round the result to the precision selected in the floating-point control register. FSDIV and FDDIV will round the result to single or double precision, respectively, regardless of the rounding precision selected in the floating-point control register.

Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	–	+	Zero	–	+	Infinity	–
In Range	+	Divide		+	inf ²	– inf ²	+	0.0	– 0.0
	–			– inf ²	+ inf ²	–	0.0	+ 0.0	
Zero	+	+ 0.0	+ 0.0	NaN ³			+	0.0	– 0.0
	–	– 0.0	+ 0.0				–	0.0	+ 0.0
Infinity	+	+ inf	– inf	+	inf	– inf	NaN‡		
	–	– inf	+ inf	–	inf	+ inf			

- NOTES:
1. If the source operand is a NaN, refer to **1.6.5 Not-A-Numbers** for more information.
 2. Sets the DZ bit in the floating-point status register exception byte.
 3. Sets the OPERR bit in the floating-point status register exception byte.

FDIV

Floating-Point Divide
(MC6888X, MC68040)

FDIV

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set for 0 ÷ 0 or infinity ÷ infinity; cleared otherwise.
	OVFL	Refer to exception processing in the appropriate user's manual.
	UNFL	Refer to exception processing in the appropriate user's manual.
	DZ	Set if the source is zero and the destination is in range; cleared otherwise.
	INEX2	Refer to exception processing in the appropriate user's manual.
	INEX1	If < fmt > is packed, refer to exception processing in the appropriate user's manual; cleared otherwise.
Accrued Exception Byte:	Affected as described in exception processing in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
											MODE			REGISTER	
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

FDIV

Floating-Point Divide (MC6888X, MC68040)

FDIV

Instruction Fields:

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, specifies the location of the source operand location. Only data addressing modes can be used as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)*

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

*This encoding in the MC68040 will cause an unimplemented data type exception to allow emulation in software.

FDIV

Floating-Point Divide (MC6888X, MC68040)

FDIV

Destination Register field—Specifies the destination floating- point data register.
Opmode field—Specifies the instruction and rounding precision.

0100000	FDIV	Rounding precision specified by the floating- point control register.
1100000	FSDIV	Single-precision rounding specified.
1100100	FDDIV	Double-precision rounding specified.

FETOX

e^x
(MC6888X, M68040FPSP)

FETOX

- Operation:** $e^{\text{Source}} \rightarrow \text{FPn}$
- Assembler** FETOX. < fmt > < ea > ,FPn
- Syntax:** FETOX.X Fm,FPn
- Syntax:** FETOX.X Fp
- Attributes:** Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates e to the power of that number. Stores the result in the destination floating-point data register.

Operation Table:

DESTINATION	SOURCE									
	+	In Range	-	+	Zero	-	+	Infinity	-	
Result	e ^x			+ 1.0			+ inf			+ 0.0

NOTE: If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.

Floating-Point Status Register:

- Condition Codes: Affected as described in **3.6.2 Conditional Testing**.
- Quotient Byte: Not affected.
- Exception Byte:
- BSUN Cleared
 - SNAN Refer to **1.6.5 Not-A-Numbers**.
 - OPERR Cleared
 - OVFL Refer to overflow in the appropriate user’s manual.
 - UNFL Refer to underflow in the appropriate user’s manual.
 - DZ Cleared
 - INEX2 Refer to inexact result in the appropriate user’s manual.
 - INEX1 If < fmt > is packed, refer to inexact result on decimal input in the appropriate user’s manual; cleared otherwise.
- Accrued Exception Byte: Affected as described in IEEE exception and trap compatibility in the appropriate user’s manual.

FETOX

e^x
(MC6888X, M68040FPSP)

FETOX

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	1	0	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

Effective Address field—Determines the addressing mode for external operands.
 If R/M = 0, this field is unused and should be all zeros.
 If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

FETOX

e^x
(MC6888X, M68040FPSP)

FETOX

R/M field—Specifies the source operand address mode.

- 0 — The operation is register to register.
- 1 — The operation is < ea > to register.

Source Specifier Field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)*
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FETOXM1

$$e^x - 1$$

(MC6888X, M68040FPSP)

FETOXM1

Operation: $e^{\text{Source}} - 1 \rightarrow \text{FPn}$

Assembler Syntax: FETOXM1. < fmt > < ea > ,FPn
FETOXM1.X Fp_m,FP_n
FETOXM1.X Fp_n

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates e to the power of that number. Subtracts one from the value and stores the result in the destination floating-point data register.

Operation Table:

DESTINATION	SOURCE							
	+	In Range	-	+	Zero	-	+	Infinity
Result		$e^x - 1$		+	0.0	-	+	inf
					- 0.0			- 1.0

NOTE: If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.

Floating-Point Status Register:

Condition Codes: Affected as described in **3.6.2 Conditional Testing**.

Quotient Byte: Not affected.

Exception Byte:

- BSUN Cleared
- SNAN Refer to **1.6.5 Not-A-Numbers**.
- OPERR Cleared
- OVFL Refer to overflow in the appropriate user’s manual.
- UNFL Refer to underflow in the appropriate user’s manual.
- DZ Cleared
- INEX2 Refer to inexact result in the appropriate user’s manual.
- INEX1 If < fmt > is packed, refer to inexact result on decimal input in the appropriate user’s manual; cleared otherwise.

Accrued Exception Byte: Affected as described in IEEE exception and trap compatibility in the appropriate user’s manual.



FETOXM1

$e^x - 1$
(MC6888X, M68040FPSP)

FETOXM1

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	1	0	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

Effective Address field—Determines the addressing mode for external operands.
If R/M = 0, this field is unused and should be all zeros.
If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

FETOXM1

$e^x - 1$
(MC6888X, M68040FPSP)

FETOXM1

R/M field—Specifies the source operand address mode.

- 0 — The operation is register to register.
- 1 — The operation is < ea > to register.

Source Specifier Field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.



FGETEXP

Get Exponent
(MC6888X, M68040FPSP)

FGETEXP

- Operation:** Exponent of Source → FPn
- Assembler Syntax:** FGETEXP.< fmt > < ea > ,FPn
FGETEXP.X FPm,FPn
FGETEXP.X FPn
- Attributes:** Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and extracts the binary exponent. Removes the exponent bias, converts the exponent to an extended-precision floating- point number, and stores the result in the destination floating- point data register.

Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	–	+	Zero	–	+	Infinity	–
Result		Exponent		+	0.0	– 0.0		NAN ²	

- NOTES:
- 1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
 - 2. Sets the OPERR bit in the floating-point status register exception byte.

Floating-Point Status Register:

- Condition Codes:** Affected as described in **3.6.2 Conditional Testing**.
- Quotient Byte:** Not affected.
- Exception Byte:**
- BSUN Cleared
 - SNAN Refer to **1.6.5 Not-A-Numbers**.
 - OPERR Set if the source is ± infinity; cleared otherwise.
 - OVFL Cleared
 - UNFL Cleared
 - DZ Cleared
 - INEX2 Cleared
 - INEX1 If < fmt > is packed, refer to inexact result on decimal input in the appropriate user’s manual; cleared otherwise.
- Accrued Exception Byte:** Affected as described in IEEE exception and trap compatibility in the appropriate user’s manual.

FGETEXP

Get Exponent
(MC6888X, M68040FPSP)

FGETEXP

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	1	1	1	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

Effective Address field—Determines the addressing mode for external operands.
 If R/M = 0, this field is unused and should be all zeros.
 If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.
 0 — The operation is register to register.
 1 — The operation is < ea > to register.

FGETEXP

Get Exponent
(MC6888X, M68040FPSP)

FGETEXP

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FGETMAN

Get Mantissa
(MC6888X, M68040FPSP)

FGETMAN

Operation: Mantissa of Source → FPn

Assembler Syntax: FGETMAN. < fmt > < ea > ,FPn
FGETMAN.X FPm,FPn
FGETMAN.X FPn

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and extracts the mantissa. Converts the mantissa to an extended-precision value and stores the result in the destination floating-point data register. The result is in the range [1.0...2.0] with the sign of the source mantissa, zero, or a NAN.

Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	–	+	Zero	–	+	Infinity	–
Result		Mantissa		+	0.0	– 0.0		NAN ²	

NOTES:

1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
2. Sets the OPERR bit in the floating-point status register exception byte.

Floating-Point Status Register:

Condition Codes: Affected as described in **3.6.2 Conditional Testing**.

Quotient Byte: Not affected.

Exception Byte:

BSUN	Cleared
SNAN	Refer to 1.6.5 Not-A-Numbers .
OPERR	Set if the source is ± infinity; cleared otherwise.
OVFL	Cleared
UNFL	Cleared
DZ	Cleared
INEX2	Cleared
INEX1	If < fmt > is packed, refer to inexact result on decimal input in the appropriate user’s manual; cleared otherwise.

Accrued Exception Byte: Affected as described in IEEE exception and trap compatibility in the appropriate user’s manual.



FGETMAN

Get Mantissa
(MC6888X, M68040FPSP)

FGETMAN

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0		SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	1	1	1

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn
An	—	—
(An)	010	reg. number:An
(An) +	011	reg. number:An
– (An)	100	reg. number:An
(d ₁₆ ,An)	101	reg. number:An
(d ₈ ,An,Xn)	110	reg. number:An
(bd,An,Xn)	110	reg. number:An
([bd,An,Xn],od)	110	reg. number:An
([bd,An],Xn,od)	110	reg. number:An

Addressing Mode	Mode	Register
(xxx).W	111	000
(xxx).L	111	001
# < data >	111	100
(d ₁₆ ,PC)	111	010
(d ₈ ,PC,Xn)	111	011
(bd,PC,Xn)	111	011
([bd,PC,Xn],od)	111	011
([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

FGETMAN

Get Mantissa
(MC6888X, M68040FPSP)

FGETMAN

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating-point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.



FINT

Integer Part (MC6888X, M68040FPSP)

FINT

- Operation:** Integer Part of Source → FPn
- Assembler Syntax:** FINT. < fmt > < ea > ,FPn
FINT.X FPm,FPn
FINT.X FPn
- Attributes:** Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary), extracts the integer part, and converts it to an extended-precision floating-point number. Stores the result in the destination floating-point data register. The integer part is extracted by rounding the extended-precision number to an integer using the current rounding mode selected in the floating-point control register mode control byte. Thus, the integer part returned is the number that is to the left of the radix point when the exponent is zero, after rounding. For example, the integer part of 137.57 is 137.0 for the round-to-zero and round-to- negative infinity modes and 138.0 for the round-to-nearest and round-to-positive infinity modes. Note that the result of this operation is a floating-point number.

Operation Table:

DESTINATION	SOURCE								
	+	In Range	–	+	Zero	–	+	Infinity	–
Result	Integer		+ 0.0		– 0.0		+ inf – inf		

NOTE: If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.

FINT

Integer Part
(MC6888X, M68040FPSP)

FINT

Floating-Point Status Register:

- Condition Codes: Affected as described in **3.6.2 Conditional Testing**.
- Quotient Byte: Not affected.
- Exception Byte:

BSUN Cleared

SNAN Refer to **1.6.5 Not-A-Numbers**.

OPERR Cleared

OVFL Cleared

UNFL Cleared

DZ Cleared

INEX2 Refer to inexact result in the appropriate user’s manual.

INEX1 If < fmt > is packed, refer to inexact result on decimal input in the appropriate user’s manual; cleared otherwise.
- Accrued Exception Byte: Affected as described in IEEE exception and trap compatibility in the appropriate user’s manual.

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID		0	0	0	EFFECTIVE ADDRESS						
											MODE	REGISTER			
0	R/M	0	SOURCE SPECIFIER		DESTINATION REGISTER			0	0	0	0	0	0	0	1

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

FINT

Integer Part (MC6888X, M68040FPSP)

FINT

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FINTRZ

Integer Part, Round-to-Zero
 (MC6888X, M68040FPSP)

FINTRZ

Operation:

Integer Part of Source → FPn

Assembler Syntax:

FINTRZ. < fmt > < ea > ,FPn
 FINTRZ.X FPm,FPn
 FINTRZ.X FPn

Attributes:

Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and extracts the integer part and converts it to an extended-precision floating-point number. Stores the result in the destination floating-point data register. The integer part is extracted by rounding the extended-precision number to an integer using the round-to-zero mode, regardless of the rounding mode selected in the floating-point control register mode control byte (making it useful for FORTRAN assignments). Thus, the integer part returned is the number that is to the left of the radix point when the exponent is zero. For example, the integer part of 137.57 is 137.0; the integer part of 0.1245 x 10² is 12.0. Note that the result of this operation is a floating-point number.

Operation Table:

DESTINATION	SOURCE								
	+	In Range	–	+	Zero	–	+	Infinity	–
Result	Integer, Forced Round-to- Zero		+ 0.0		– 0.0		+ inf		– inf

NOTE: If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.



FINTRZ

Integer Part, Round-to-Zero
(MC6888X, M68040FPSP)

FINTRZ

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Cleared
	OVFL	Cleared
	UNFL	Cleared
	DZ	Cleared
	INEX2	Refer to inexact result in the appropriate user's manual.
	INEX1	If < fmt > is packed, refer to inexact result on decimal input in the appropriate user's manual; cleared otherwise.
Accrued Exception Byte:	Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	0	0	1	1

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

FINTRZ

Integer Part, Round-to-Zero (MC6888X, M68040FPSP)

FINTRZ

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If RM = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FLOG10

Log₁₀
(MC6888X, M68040FPSP)

FLOG10

- Operation:** Log₁₀ of Source → FPn
- Assembler Syntax:** FLOG10. < fmt > < ea > ,FPn
FLOG10.X FPm,FPn
FLOG10.X FPn
- Attributes:** Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Convert the source operand to extended precision (if necessary) and calculates the logarithm of that number using base 10 arithmetic. Stores the result in the destination floating-point data register. This function is not defined for input values less than zero.

Operation Table:

DESTINATION	SOURCE ¹					
	+	In Range	–	+	Zero	–
Result	Log ₁₀	NAN ²		–	inf ³	
				+	inf	NAN ²

- NOTES:
1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
 2. Sets the OPERR bit in the floating-point status register exception byte.
 3. Sets the DZ bit in the floating-point status register exception byte.

Floating-Point Status Register:

- Condition Codes:** Affected as described in **3.6.2 Conditional Testing**.
- Quotient Byte:** Not affected.
- Exception Byte:**
 - BSUN Cleared
 - SNAN Refer to **1.6.5 Not-A-Numbers**.
 - OPERR Set if the source operand is < 0; cleared otherwise.
 - OVFL Cleared
 - UNFL Cleared
 - DZ Set if the source is ± 0; cleared otherwise
 - INEX2 Refer to inexact result in the appropriate user’s manual.
 - INEX1 If < fmt > is packed, refer to inexact result on decimal input in the appropriate user’s manual; cleared otherwise.
- Accrued Exception Byte:** Affected as described in IEEE exception and trap compatibility in the appropriate user’s manual.

FLOG10

Log₁₀
(MC6888X, M68040FPSP)

FLOG10

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	0	1	0	1

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

Effective Address field—Determines the addressing mode for external operands.
 If R/M = 0, this field is unused and should be all zeros.
 If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

FLOG10

Log₁₀
(MC6888X, M68040FPSP)

FLOG10

R/M field—Specifies the source operand address mode.

- 0 — The operation is register to register.
- 1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FLOG2

Log₂
(MC6888X, M68040FPSP)

FLOG2

Operation: Log₂ of Source → FPn

Assembler Syntax: FLOG2. < fmt > < ea > ,FPn
FLOG2.X FPm,FPn
FLOG2.X FPn

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates the logarithm of that number using base two arithmetic. Stores the result in the destination floating- point data register. This function is not defined for input values less than zero.

Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	–	+	Zero	–	+	Infinity	–
Result	Log ₂ NAN ²			– inf ³			+ inf NAN ²		

- NOTES:
1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
 2. Sets the OPERR bit in the floating-point status register exception byte.
 3. Sets the DZ bit in the floating-point status register exception byte.

Floating-Point Status Register:

Condition Codes: Affected as described in **3.6.2 Conditional Testing**.

Quotient Byte: Not affected.

Exception Byte:

BSUN	Cleared
SNAN	Refer to 1.6.5 Not-A-Numbers .
OPERR	Set if the source is < 0; cleared otherwise
OVFL	Cleared
UNFL	Cleared
DZ	Set if the source is ± 0; cleared otherwise
INEX2	Refer to inexact result in the appropriate user's manual.
INEX1	If < fmt > is packed, refer to inexact result on decimal input in the appropriate user's manual; cleared otherwise.

Accrued Exception Byte: Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.



FLOG2

Log₂
(MC6888X, M68040FPSP)

FLOG2

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	0	1	1	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

Effective Address field—Determines the addressing mode for external operands.
 If R/M = 0, this field is unused and should be all zeros.
 If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

FLOG2

Log₂
(MC6888X, M68040FPSP)

FLOG2

R/M field—Specifies the source operand address mode.

- 0 — The operation is register to register.
- 1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.



FLOGN

Log_e
(MC6888X, M68040FPSP)

FLOGN

- Operation:** Log_e of Source → FPn
- Assembler Syntax:** FLOGN. < fmt > < ea > ,FPn
FLOGN.X FPM,FPn
FLOGN.X FPn
- Attributes:** Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates the natural logarithm of that number. Stores the result in the destination floating-point data register. This function is not defined for input values less than zero.

Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	–	+	Zero	–	+	Infinity	–
Result	ln(x)		NaN ²		– inf ³		+ inf		NaN ²

- NOTES:
- 1. If the source operand is a NaN, refer to **1.6.5 Not-A-Numbers** for more information.
 - 2. Sets the OPERR bit in the floating-point status register exception byte.
 - 3. Sets the DZ bit in the floating-point status register exception byte.

Floating-Point Status Register:

- Condition Codes:** Affected as described in **3.6.2 Conditional Testing**.
- Quotient Byte:** Not affected.
- Exception Byte:**
- | | |
|-------|---|
| BSUN | Cleared |
| SNAN | Refer to 1.6.5 Not-A-Numbers . |
| OPERR | Set if the source operand is < 0; cleared otherwise. |
| OVFL | Cleared |
| UNFL | Cleared |
| DZ | Set if the source is ± 0; cleared otherwise |
| INEX2 | Refer to inexact result in the appropriate user’s manual. |
| INEX1 | If < fmt > is packed, refer to inexact result on decimal input in the appropriate user’s manual; cleared otherwise. |
- Accrued Exception Byte:** Affected as described in IEEE exception and trap compatibility in the appropriate user’s manual.

FLOGN

Log_e
(MC6888X, M68040FPSP)

FLOGN

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	0	1	0	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

Effective Address field—Determines the addressing mode for external operands.
 If R/M = 0, this field is unused and should be all zeros.
 If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.



FLOGN

Log_e
(MC6888X, M68040FPSP)

FLOGN

R/M field—Specifies the source operand address mode.

- 0 — The operation is register to register.
- 1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FLOGNP1

$\text{Log}_e (x + 1)$
(MC6888X, M68040FPSP)

FLOGNP1

Operation:

Log_e of (Source + 1) → FPn

Assembler

FLOGNP1. < fmt > < ea > ,FPn

Syntax:

FLOGNP1.X FPm,FPn
FLOGNP1.X FPn

Attributes:

Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description:

Converts the source operand to extended precision (if necessary), adds one to that value, and calculates the natural logarithm of that intermediate result. Stores the result in the destination floating-point data register. This function is not defined for input values less than − 1.

Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	−	+	Zero	−	+	Infinity	−
Result	ln(x + 1)	ln(x + 1) ²	+ 0.0		− 0.0		+ inf		NAN ²³

- NOTES:
1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
 2. If the source is − 1, sets the DZ bit in the floating-point status register exception byte and returns a NAN. If the source is < − 1, sets the OPERR bit in the floating-point status register exception byte and returns a NAN.
 3. Sets the OPERR bit in the floating-point status register exception byte.

Floating-Point Status Register:

Condition Codes:

Affected as described in **3.6.2 Conditional Testing**.

Quotient Byte:

Not affected.



FLOGNP1

$\text{Log}_e (x + 1)$
(MC6888X, M68040FPSP)

FLOGNP1

Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the source operand is < -1 ; cleared otherwise.
	OVFL	Cleared
	UNFL	Refer to underflow in the appropriate user's manual.
	DZ	Set if the source operand is -1 ; cleared otherwise
	INEX2	Refer to inexact result in the appropriate user's manual.
	INEX1	If $< \text{fmt} >$ is packed, refer to inexact result on decimal input in the appropriate user's manual; cleared otherwise.

Accrued Exception Byte: Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	0	1	1	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

FLOGNP1

$\text{Log}_e (x + 1)$
(MC6888X, M68040FPSP)

FLOGNP1

Effective Address field—Determines the addressing mode for external operands.
 If R/M = 0, this field is unused and should be all zeros.
 If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.
 0 — The operation is register to register.
 1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.
 If R/M = 0, specifies the source floating-point data register.
 If R/M = 1, specifies the source data format:
 000 — Long-Word Integer (L)
 001 — Single-Precision Real (S)
 010 — Extended-Precision Real (X)
 011 — Packed-Decimal Real (P)
 100 — Word Integer (W)
 101 — Double-Precision Real (D)
 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FMOD

Modulo Remainder (MC6888X, M68040FPSP)

FMOD

Operation: Modulo Remainder of (FPn ÷ Source) → FPn

Assembler FMOD. < fmt > < ea > ,FPn

Syntax: FMOD.X FPM,FPn

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates the modulo remainder of the number in the destination floating-point data register, using the source operand as the modulus. Stores the result in the destination floating-point data register and stores the sign and seven least significant bits of the quotient in the floating-point status register quotient byte (the quotient is the result of FPn ÷ Source). The modulo remainder function is defined as:

$$FPn - (Source \times N)$$

where N = INT(FPn ÷ Source) in the round-to-zero mode.

The FMOD function is not defined for a source operand equal to zero or for a destination operand equal to infinity. Note that this function is not the same as the FREM instruction, which uses the round-to-nearest mode and thus returns the remainder that is required by the IEEE *Specification for Binary Floating-Point Arithmetic*.

Operation Table:

DESTINATION		SOURCE ¹								
		+	In Range	−	+	Zero#	−	+	Infinity	−
In Range	+ −		Modulo Remainder			NAN ²			FPn ³	
Zero	+ −		+ 0.0 − 0.0			NAN ²			+ 0.0 − 0.0	
Infinity	+ −		NAN ²			NAN ²			NAN ²	

NOTES:

1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
2. Sets the OPERR bit in the floating-point status register exception byte.
3. Returns the value of FPn before the operation. However, the result is processed by the normal instruction termination procedure to round it as required. Thus, an overflow and/or inexact result may occur if the rounding precision has been changed to a smaller size since the FPn value was loaded

FMOD

Modulo Remainder
(MC6888X, M68040FPSP)

FMOD

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Loaded with the sign and least significant seven bits of the quotient ($FPn \div \text{Source}$). The sign of the quotient is the exclusive-OR of the sign bits of the source and destination operands.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the source is zero or the destination is infinity; cleared otherwise.
	OVFL	Cleared
	UNFL	Refer to underflow in the appropriate user's manual.
	DZ	Cleared
	INEX2	Refer to inexact result in the appropriate user's manual.
	INEX1	If <code>< fmt ></code> is packed, in the appropriate user's manual for inexact result on decimal input; cleared otherwise.
Accrued Exception Byte:	Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	1	0	1	1	0	1

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

FMOD

Modulo Remainder (MC6888X, M68040FPSP)

FMOD

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register.

FMOVE

Move Floating-Point Data Register (MC6888X, MC68040)

FMOVE

Operation: Source → Destination

Assembler FMOVE. < fmt > < ea > ,FPn

Syntax: FMOVE. < fmt > FPm, < ea >

FMOVE.P FPm, < ea > {Dn}

FMOVE.P FPm, < ea > {k}

*FrMOVE. < fmt > < ea > ,FPn

where r is rounding precision, S or D

*Supported by MC68040 only

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Moves the contents of the source operand to the destination operand. Although the primary function of this instruction is data movement, it is also considered an arithmetic instruction since conversions from the source operand format to the destination operand format are performed implicitly during the move operation. Also, the source operand is rounded according to the selected rounding precision and mode.

Unlike the MOVE instruction, the FMOVE instruction does not support a memory-to-memory format. For such transfers, it is much faster to utilize the MOVE instruction to transfer the floating-point data than to use the FMOVE instruction. The FMOVE instruction only supports memory-to-register, register-to-register, and register-to-memory operations (in this context, memory may refer to an integer data register if the data format is byte, word, long, or single). The memory-to-register and register-to-register operation uses a command word encoding distinctly different from that used by the register-to-memory operation; these two operation classes are described separately.

Memory-to-Register and Register-to-Register Operation: Converts the source operand to an extended-precision floating-point number (if necessary) and stores it in the destination floating-point data register. MOVE will round the result to the precision selected in the floating-point control register. FMOVE and FDMOVE will round the result to single or double precision, respectively, regardless of the rounding precision selected in the floating-point control register. Depending on the source data format and the rounding precision, some operations may produce an inexact result. In the following table, combinations that can produce an inexact result are marked with a dot (·), but all other combinations produce an exact result.



FMOVE

Move Floating-Point Data Register
(MC6888X, MC68040)

FMOVE

Rounding Precision	Source Format						
	B	W	L	S	D	X	P
Single			.		.	.	.
Double						.	.
Extended							.

Floating-Point Status Register (< ea > to Register):

- Condition Codes:Affected as described in 3.6.2 Conditional Testing.
- Quotient Byte:Not affected.
- Exception Byte:

BSUNCleared

SNANRefer to 1.6.5 Not-A-Numbers.

OPERRCleared

OVFLCleared

UNFLRefer to exception processing in the appropriate user’s manual if the source is an extended-precision denormalized number; cleared otherwise.

DZCleared

INEX2Refer to exception processing in the appropriate user’s manual if < fmt > is L,D, or X; cleared otherwise.

INEX1Refer to exception processing in the appropriate user’s manual if < fmt > is P; cleared otherwise.
- Accrued Exception Byte:Affected as described in exception processing in the appropriate user’s manual.

FMOVE

Move Floating-Point Data Register

(MC6888X, MC68040)

FMOVE

Instruction Format:

< EA > TO REGISTER

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS MODE REGISTER					
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

Instruction Fields:

Effective Address field—Determines the addressing mode for external operands.
 If R/M = 0, this field is unused and should be all zeros.
 If R/M = 1, specifies the location of the source operand. Only data addressing modes can be used as listed in the following table:

Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn
An	—	—
(An)	010	reg. number:An
(An) +	011	reg. number:An
– (An)	100	reg. number:An
(d ₁₆ ,An)	101	reg. number:An
(d ₈ ,An,Xn)	110	reg. number:An
(bd,An,Xn)	110	reg. number:An
([bd,An,Xn],od)	110	reg. number:An
([bd,An],Xn,od)	110	reg. number:An

Addressing Mode	Mode	Register
(xxx).W	111	000
(xxx).L	111	001
# < data >	111	100
(d ₁₆ ,PC)	111	010
(d ₈ ,PC,Xn)	111	011
(bd,PC,Xn)	111	011
([bd,PC,Xn],od)	111	011
([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.
 0 — The operation is register to register.
 1 — The operation is < ea > to register.

FMOVE

Move Floating-Point Data Register (MC6888X, MC68040)

FMOVE

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)*
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

*This encoding in the MC68040 will cause an unimplemented data type exception to allow emulation in software.

Destination Register field—Specifies the destination floating- point data register.

Opmode field—Specifies the instruction and rounding precision.

- 0000000 FMOVE Rounding precision specified by the floating-point control register.
- 1000000 FSMOVE Single-precision rounding specified.
- 1000100 FDMOVE Double-precision rounding specified.

Register-to-Memory Operation: Rounds the source operand to the size of the specified destination format and stores it at the destination effective address. If the format of the destination is packed decimal, a third operand is required to specify the format of the resultant string. This operand, called the k-factor, is a 7-bit signed integer (twos complement) and may be specified as an immediate value or in an integer data register. If a data register contains the k-factor, only the least significant seven bits are used, and the rest of the register is ignored.

FMOVE

Move Floating-Point Data Register (MC6888X, MC68040)

FMOVE

Floating-Point Status Register (Register-to-Memory):

Condition Codes:	Not affected.	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Cleared
< fmt > is B, W, or L	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the source operand is infinity or if the destination size is exceeded after conversion and rounding; cleared otherwise.
	OVFL	Cleared
	UNFL	Cleared
	DZ	Cleared
	INEX2	Refer to exception processing in the appropriate user's manual.
	INEX1	Cleared
	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers
	OVFL	Refer to exception processing in the appropriate user's manual.
< fmt > is S, D, or X	UNFL	Refer to exception processing in the appropriate user's manual.
	DZ	Cleared
	INEX2	Refer to exception processing in the appropriate user's manual.
	INEX1	Cleared
	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the k-factor > + 17 or the magnitude of the decimal exponent exceeds three digits; cleared otherwise.
	OVFL	Cleared
	UNFL	Cleared
	DZ	Cleared
< fmt > is P	INEX2	Refer to exception processing in the appropriate user's manual.
	INEX1	Cleared
	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the k-factor > + 17 or the magnitude of the decimal exponent exceeds three digits; cleared otherwise.
	OVFL	Cleared
	UNFL	Cleared
	DZ	Cleared
	INEX2	Refer to exception processing in the appropriate user's manual.
	INEX1	Cleared
Accrued Exception Byte:	Affected as described in exception processing in the appropriate user's manual.	



FMOVE

Move Floating-Point Data Register (MC6888X, MC68040)

FMOVE

Instruction Format:

REGISTER—TO-MEMORY

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	COPROCESSOR ID			1	0	0	0	EFFECTIVE ADDRESS					
											MODE			REGISTER	
0	1	1	DESTINATION FORMAT			SOURCE REGISTER			K-FACTOR (IF REQUIRED)						

Instruction Fields:

Effective Address field—Specifies the destination location. Only data alterable addressing modes can be used as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	—	—
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	—	—
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	—	—
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	—	—
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	—	—
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	—	—

*Only if < fmt > is byte, word, long, or single.

Destination Format field—Specifies the data format of the destination operand:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real with Static k-Factor (P{#k})*
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)
- 111 — Packed-Decimal Real with Dynamic k-Factor (P{Dn})*

*This encoding will cause an unimplemented data type exception in the MC68040 to allow emulation in software.

FMOVE

Move Floating-Point Data Register
(MC6888X, MC68040)

FMOVE

Source Register field—Specifies the source floating-point data register.

k-Factor field—If the destination format is packed decimal, used to specify the format of the decimal string. For any other destination format, this field should be set to all zeros. For a static k-factor, this field is encoded with a twos-complement integer where the value defines the format as follows:

- 64 to 0—Indicates the number of significant digits to the right of the decimal point (FORTRAN "F" format).
- + 1 to + 17—Indicates the number of significant digits in the mantissa (FORTRAN "E" format).
- + 18 to + 63—Sets the OPERR bit in the floating-point status register exception byte and treated as + 17.

The format of this field for a dynamic k-factor is:

rrr0000

where "rrr" is the number of the main processor data register that contains the k-factor value.

The following table gives several examples of how the k-factor value affects the format of the decimal string that is produced by the floating-point coprocessor. The format of the string that is generated is independent of the source of the k-factor (static or dynamic).

k- Factor	Source Operand Value	Destination String
– 5	+ 12345.678765	+ 1.234567877E + 4
– 3	+ 12345.678765	+ 1.2345679E + 4
– 1	+ 12345.678765	+ 1.23457E + 4
0	+ 12345.678765	+ 1.2346E + 4
+ 1	+ 12345.678765	+ 1.E + 4
+ 3	+ 12345.678765	+ 1.23E + 4
+ 5	+ 12345.678765	+ 1.2346E + 4



FMOVE

Move Floating-Point System Control Register (MC6888X, MC68040)

FMOVE

Operation: Source → Destination

Assembler Syntax: FMOVE.L < ea > ,FPCR
FMOVE.L FPCR, < ea >

Attributes: Size = (Long)

Description: Moves the contents of a floating-point system control register (floating-point control register, floating-point status register, or floating-point instruction address register) to or from an effective address. A 32-bit transfer is always performed, even though the system control register may not have 32 implemented bits. Unimplemented bits of a control register are read as zeros and are ignored during writes (must be zero for compatibility with future devices). For the MC68881, this instruction does not cause pending exceptions (other than protocol violations) to be reported. Furthermore, a write to the floating-point control register exception enable byte or the floating-point status register exception status byte cannot generate a new exception, regardless of the value written.

Floating-Point Status Register: Changed only if the destination is the floating-point status register, in which case all bits are modified to reflect the value of the source operand.

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID		0	0	0	EFFECTIVE ADDRESS						
											MODE		REGISTER		
1	0	dr	REGISTER SELECT			0	0	0	0	0	0	0	0	0	0

FMOVE

Move Floating-Point System Control Register (MC6888X, MC68040)

FMOVE

Instruction Fields:

Effective Address field—(Memory-to-Register) All addressing modes can be used as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn	000	reg. number:Dn	(xxx).W	111	000
An*	001	reg. number:An	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if the source register is the floating-point instruction address register.

Effective Address field—(Register-to-Memory) Only alterable addressing modes can be used as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn	000	reg. number:Dn	(xxx).W	111	000
An*	001	reg. number:An	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	—	—
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	—	—
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	—	—
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	—	—
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	—	—
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	—	—

*Only if the source register is the floating-point instruction address register.



FMOVE

Move Floating-Point System Control Register (MC6888X, MC68040)

FMOVE

dr field—Specifies the direction of the data transfer.

0 — From < ea > to the specified system control register.

1 — From the specified system control register to < ea > .

Register Select field—Specifies the system control register to be moved:

100 Floating-Point Control Register

010 Floating-Point Status Register

001 Floating-Point Instruction Address Register

FMOVECR

Move Constant ROM
(MC6888X, M68040FPSP)

FMOVECR

Operation: ROM Constant → FPn

Assembler

Syntax: FMOVECR.X # < ccc > ,FPn

Attributes: Format = (Extended)

Description: Fetches an extended-precision constant from the floating- point coprocessor on-chip ROM, rounds the mantissa to the precision specified in the floating-point control register mode control byte, and stores it in the destination floating-point data register. The constant is specified by a predefined offset into the constant ROM. The values of the constants contained in the ROM are shown in the offset table at the end of this description.

Floating-Point Status Register:

- Condition Codes: Affected as described in **3.6.2 Conditional Testing**.
- Quotient Byte: Not affected.
- Exception Byte:
 - BSUN Cleared
 - SNAN Cleared
 - OPERR Cleared
 - OVFL Cleared
 - UNFL Cleared
 - DZ Cleared
 - INEX2 Refer to inexact result in the appropriate user's manual.
 - INEX1 Cleared
- Accrued Exception Byte: Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	0	0	0	0	0	0
0	1	0	1	1	1	DESTINATION REGISTER			ROM OFFSET						



FMOVECR

Move Constant ROM
(MC6888X, M68040FPSP)

FMOVECR

Instruction Fields:

- Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.
- Destination Register field—Specifies the destination floating- point data register.
- ROM Offset field—Specifies the offset into the floating-point coprocessor on-chip constant ROM where the desired constant is located. The offsets for the available constants are as follows:

Offset	Constant
\$00	π
\$0B	$\text{Log}_{10}(2)$
\$0C	e
\$0D	$\text{Log}_2(e)$
\$0E	$\text{Log}_{10}(e)$
\$0F	0.0
\$30	$\ln(2)$
\$31	$\ln(10)$
\$32	100
\$33	10^1
\$34	10^2
\$35	10^4
\$36	10^8
\$37	10^{16}
\$38	10^{32}
\$39	10^{64}
\$3A	10^{128}
\$3B	10^{256}
\$3C	10^{512}
\$3D	10^{1024}
\$3E	10^{2048}
\$3F	10^{4096}

The on-chip ROM contains other constants useful only to the on- chip microcode routines. The values contained at offsets other than those defined above are reserved for the use of Motorola and may be different on various mask sets of the floating-point coprocessor. These undefined values yield the value 0.0 in the M68040FPSP.

FMOVEM

Move Multiple Floating-Point Data Registers (MC6888X, MC68040)

FMOVEM

Operation: Register List → Destination
Source → Register List

Assembler Syntax: FMOVEM.X < list > , < ea >
FMOVEM.X Dn, < ea >
FMOVEM.X < ea > , < list > FMOVEM.X < ea > ,Dn

Attributes: Format = (Extended)

Description: Moves one or more extended-precision numbers to or from a list of floating-point data registers. No conversion or rounding is performed during this operation, and the floating-point status register is not affected by the instruction. For the MC68881, this instruction does not cause pending exceptions (other than protocol violations) to be reported. Furthermore, a write to the floating-point control register exception enable byte or the floating-point status register exception status byte cannot generate a new exception, despite the value written.

Any combination of the eight floating-point data registers can be transferred, with the selected registers specified by a user-supplied mask. This mask is an 8-bit number, where each bit corresponds to one register; if a bit is set in the mask, that register is moved. The register select mask may be specified as a static value contained in the instruction or a dynamic value in the least significant eight bits of an integer data register (the remaining bits of the register are ignored).

FMOVEM allows three types of addressing modes: the control modes, the predecrement mode, or the postincrement mode. If the effective address is one of the control addressing modes, the registers are transferred between the processor and memory starting at the specified address and up through higher addresses. The order of the transfer is from FP0 – FP7.



FMOVEM

Move Multiple Floating-Point
Data Registers
(MC6888X, MC68040)

FMOVEM

If the effective address is the predecrement mode, only a register- to-memory operation is allowed. The registers are stored starting at the address contained in the address register and down through lower addresses. Before each register is stored, the address register is decremented by 12 (the size of an extended-precision number in memory) and the floating-point data register is then stored at the resultant address. When the operation is complete, the address register points to the image of the last floating- point data register stored. The order of the transfer is from FP7 – FP0.

If the effective address is the postincrement mode, only a memory- to-register operation is allowed. The registers are loaded starting at the specified address and up through higher addresses. After each register is stored, the address register is incremented by 12 (the size of an extended-precision number in memory). When the operation is complete, the address register points to the byte immediately following the image of the last floating-point data register loaded. The order of the transfer is the same as for the control addressing modes: FP0 – FP7.

Floating-Point Status Register: Not Affected. Note that the FMOVEM instruction provides the only mechanism for moving a floating- point data item between the floating-point unit and memory without performing any data conversions or affecting the condition code and exception status bits.

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
											MODE			REGISTER	
1	1	dr	MODE		0	0	0	REGISTER LIST							

FMOVEM

Move Multiple Floating-Point Data Registers (MC6888X, MC68040)

FMOVEM

Instruction Fields:

Effective Address field—(Memory-to-Register) Only control addressing modes or the postincrement addressing mode can be used as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn	—	—	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	—	—
(An) +	011	reg. number:An			
– (An)	—	—			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

Effective Address field—(Register-to-Memory) Only control alterable addressing modes or the predecrement addressing mode can be used as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn	—	—	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	—	—
(An) +	—	—			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	—	—
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	—	—
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	—	—
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	—	—
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	—	—

FMOVEM

Move Multiple Floating-Point Data Registers (MC6888X, MC68040)

FMOVEM

dr field—Specifies the direction of the transfer.

- 0 — Move the listed registers from memory to the floating-point unit.
- 1 — Move the listed registers from the floating-point unit to memory.

Mode field—Specifies the type of the register list and addressing mode.

- 00 — Static register list, predecrement addressing mode.
- 01 — Dynamic register list, predecrement addressing mode.
- 10 — Static register list, postincrement or control addressing mode.
- 11 — Dynamic register list, postincrement or control addressing mode.

Register List field:

Static list—contains the register select mask. If a register is to be moved, the corresponding bit in the mask is set as shown below; otherwise it is clear.

Dynamic list—contains the integer data register number, rrr, as listed in the following table:

List Type	Register List Format							
Static, – (An)	FP7	FP6	FP5	FP4	FP3	FP2	FP1	FP0
Static, (An) + , or Control	FP0	FP1	FP2	FP3	FP4	FP5	FP6	FP7
Dynamic	0	r	r	r	0	0	0	0

The format of the dynamic list mask is the same as for the static list and is contained in the least significant eight bits of the specified main processor data register.

Programming Note: This instruction provides a very useful feature, dynamic register list specification, that can significantly enhance system performance. If the calling conventions used for procedure calls utilize the dynamic register list feature, the number of floating-point data registers saved and restored can be reduced.

To utilize the dynamic register specification feature of the FMOVEM instruction, both the calling and the called procedures must be written to communicate information about register usage. When one procedure calls another, a register mask must be passed to the called procedure to indicate which registers must not be altered upon return to the calling procedure. The called procedure then saves only those registers that are modified and are already in use. Several techniques can be used to utilize this mechanism, and an example follows.

FMOVE

Move Multiple Floating-Point Data Registers (MC6888X, MC68040)

FMOVE

In this example, a convention is defined by which each called procedure is passed a word mask in D7 that identifies all floating-point registers in use by the calling procedure. Bits 15 – 8 identify the registers in the order FP0 – FP7, and bits 7 – 0 identify the registers in the order FP7 – FP0 (the two masks are required due to the different transfer order used by the predecrement and postincrement addressing modes). The code used by the calling procedure consists of simply moving the mask (which is generated at compile time) for the floating-point data registers currently in use into D7:

Calling procedure...

MOVE.W	#ACTIVE,D7	Load the list of FP registers that are in use.
BSR	PROC_2	

The entry code for all other procedures computes two masks. The first mask identifies the registers in use by the calling procedure that are used by the called procedure (and therefore saved and restored by the called procedure). The second mask identifies the registers in use by the calling procedure that are used by the called procedure (and therefore not saved on entry). The appropriate registers are then stored along with the two masks:

Called procedure...

MOVE.W	D7,D6	Copy the list of active registers.
AND.W	#WILL_USE,D7	Generate the list of doubly-used registers.
FMOVE	D7, – (A7)	Save those registers.
MOVE.W	D7, – (A7)	Save the register list.
EOR.W	D7,D6	Generate the list of not saved active registers.
MOVE.W	D6, – (A7)	Save it for later use.

If the second procedure calls a third procedure, a register mask is passed to the third procedure that indicates which registers must not be altered by the third procedure. This mask identifies any registers in the list from the first procedure that were not saved by the second procedure, plus any registers used by the second procedure that must not be altered by the third procedure.



FMOVEM

Move Multiple Floating-Point
Data Registers
(MC6888X, MC68040)

FMOVEM

An example of the calculation of this mask is as follows:

Nested calling sequence...

MOVE.W	UNSAVED (A7),D7	Load the list of active registers not saved at entry.
OR.W	#WILL_USE,D7	Combine with those active at this time
BSR	PROC_3	

Upon return from a procedure, the restoration of the necessary registers follows the same convention, and the register mask generated during the save operation on entry is used to restore the required floating-point data registers:

Return to caller...

ADDQ.L	#2,A7	Discard the list of registers not saved.
MOVE.B	(A7) + ,D7	Get the saved register list (pop word, use byte).
FMOVEM	(A7) + ,D7	Restore the registers.
	*	
	*	
	*	
RTS		Return to the calling routine.

FMOVEM

Move Multiple Floating-Point Control Registers (MC6888X, MC68040)

FMOVEM

Operation: Register List → Destination
Source → Register List

Assembler Syntax: FMOVEM.L < list > , < ea >
FMOVEM.L < ea > , < list >

Attributes: Size = (Long)

Description: Moves one or more 32-bit values into or out of the specified system control registers. Any combination of the three system control registers may be specified. The registers are always moved in the same order, regardless of the addressing mode used; the floating-point control register is moved first, followed by the floating-point status register, and the floating-point instruction address register is moved last. If a register is not selected for the transfer, the relative order of the transfer of the other registers is the same. The first register is transferred between the floating-point unit and the specified address, with successive registers located up through higher addresses.

For the MC68881, this instruction does not cause pending exceptions (other than protocol violations) to be reported. Furthermore, a write to the floating-point control register exception enable byte or the floating-point status register exception status byte cannot generate a new exception, despite the value written.

When more than one register is moved, the memory or memory- alterable addressing modes can be used as shown in the addressing mode tables. If the addressing mode is predecrement, the address register is first decremented by the total size of the register images to be moved (i.e., four times the number of registers), and then the registers are transferred starting at the resultant address. For the postincrement addressing mode, the selected registers are transferred to or from the specified address, and then the address register is incremented by the total size of the register images transferred. If a single system control register is selected, the data register direct addressing mode may be used; if the only register selected is the floating-point instruction address register, then the address register direct addressing mode is allowed. Note that if a single register is selected, the opcode generated is the same as for the FMOVE single system control register instruction.

FMOVEM

Move Multiple Floating-Point Control Registers (MC6888X, MC68040)

FMOVEM

Floating-Point Status Register: Changed only if the destination list includes the floating-point status register in which case all bits are modified to reflect the value of the source register image.

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
1	0	dr	REGISTER LIST			0	0	0	0	0	0	0	0	0	0

Instruction Fields:

Effective Address field—Determines the addressing mode for the operation.

Memory-to-Register—Only control addressing modes or the postincrement addressing mode can be used as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An**	001	reg. number:An	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if a single floating-point instruction address register, floating-point status register, or floating-point control register is selected.
 **Only if the floating-point instruction address register is the single register selected.

FMOVEM

Move Multiple Floating-Point Control Registers (MC6888X, MC68040)

FMOVEM

Register-to-Memory—Only control alterable addressing modes or the predecrement addressing mode can be used as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An**	001	reg. number:An	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	—	—
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	—	—
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	—	—
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	—	—
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	—	—
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	—	—

*Only if a single floating-point control register is selected.
 **Only if the floating-point instruction address register is the single register selected.

dr field—Specifies the direction of the transfer.
 0 — Move the listed registers from memory to the floating-point unit.
 1 — Move the listed registers from the floating-point unit to memory.

Register List field—Contains the register select mask. If a register is to be moved, the corresponding bit in the list is set; otherwise, it is clear. At least one register must be specified.

Bit Number	Register
12	Floating-Point Control Register
11	Floating-Point Status Register
10	Floating-Point Instruction Address Register

FMUL

Floating-Point Multiply (MC6888X, MC68040)

FMUL

Operation: Source x FPn → FPn

Assembler Syntax: FMUL. < fmt > < ea > ,FPn
 FMUL.X FPM,FPn
 *FrMUL < fmt > < ea > ,FPn
 *FrMUL.X FPM,FPn
 where r is rounding precision, S or D
 *Supported by MC68040 only

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and multiplies that number by the number in the destination floating-point data register. Stores the result in the destination floating-point data register.

FMUL will round the result to the precision selected in the floating-point control register. FSMUL and FDMUL will round the result to single or double precision, respectively, regardless of the rounding precision selected in the floating-point control register.

Operation Table:

DESTINATION	SOURCE ¹					
		In Range	Zero	Infinity		
In Range	+	Multiply	+ 0.0	– 0.0	+ inf	– inf
	–		– 0.0	+ 0.0	– inf	+ inf
Zero	+	+ 0.0	– 0.0	+ 0.0	NaN ²	
	–	– 0.0	+ 0.0	– 0.0		
Infinity	+	+ inf	– inf	NaN ²		– inf
	–	– inf	+ inf			+ inf

- NOTES:
1. If the source operand is a NaN, refer to **1.6.5 Not-A-Numbers** for more information.
 2. Sets the OPERR bit in the floating-point status register exception byte.

FMUL

Floating-Point Multiply

(MC6888X, MC68040)

FMUL

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set for 0 x infinity; cleared otherwise.
	OVFL	Refer to exception processing in the appropriate user's manual.
	UNFL	Refer to exception processing in the appropriate user's manual.
	DZ	Cleared
	INEX2	Refer to exception processing in the appropriate user's manual.
	INEX1	If < fmt > is packed, refer to exception processing in the appropriate user's manual; cleared otherwise.
Accrued Exception Byte:	Affected as described in exception processing in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS MODE REGISTER					
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

FMUL

Floating-Point Multiply (MC6888X, MC68040)

FMUL

Instruction Fields:

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, specifies the location of the source operand location. Only data addressing modes can be used as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)*

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

*This encoding will cause an unimplemented data type exception in the MC68040 to allow emulation in software.

FMUL

Floating-Point Multiply (MC6888X, MC68040)

FMUL

Destination Register field—Specifies the destination floating- point data register.

Opmode field—Specifies the instruction and rounding precision.

0100011	FMUL	Rounding precision specified by the floating-point control register.
1100011	FSMUL	Single-precision rounding specified.
1100111	FDMUL	Double-precision rounding specified.



FNEG

Floating-Point Negate (MC6888X, MC68040)

FNEG

Operation: $-(\text{Source}) \rightarrow \text{FPn}$

Assembler Syntax: FNEG. < fmt > < ea > ,FPn
FNEG.X FPM,FPn
FNEG.X FPN
*FrNEG. < fmt > < ea > ,FPn
*FrNEG.X FPM,FPn
*FrNEG.X FPN
where r is rounding precision, S or D
*Supported by MC68040 only

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and inverts the sign of the mantissa. Stores the result in the destination floating-point data register.

FNEG will round the result to the precision selected in the floating-point control register. FSNEG and FDNEG will round the result to single or double precision, respectively, regardless of the rounding precision selected in the floating-point control register.

Operation Table:

DESTINATION	SOURCE					
	+	In Range	-	+	Zero	-
Result		Negate	-	0.0	+ 0.0	- inf

NOTE: If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.

FNEG

Floating-Point Negate (MC6888X, MC68040)

FNEG

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Cleared
	OVFL	Cleared
	UNFL	If source is an extended-precision denormalized number, refer to exception processing in the appropriate user's manual; cleared otherwise.
	DZ	Cleared
	INEX2	Cleared
Accrued Exception Byte:		Affected as described in exception processing in the appropriate user's manual.
INEX1		If < fmt > is packed, refer to exception processing in the appropriate user's manual; cleared otherwise.

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS MODEREGISTER					
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

FNEG

Floating-Point Negate (MC6888X, MC68040)

FNEG

Instruction Fields:

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, specifies the location of the source operand. Only data addressing modes can be used as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)*

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

*This encoding will cause an unimplemented data type exception to allow emulation in software.

FNEG

Floating-Point Negate
(MC6888X, MC68040)

FNEG

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

Opmode field—Specifies the instruction and rounding precision.

0011010	FNEG	Rounding precision specified by the floating-point control register.
1011010	FSNEG	Single-precision rounding specified.
1011110	FDNEG	Double-precision rounding specified.

FNOP

No Operation
(MC6888X, MC68040)

FNOP

Operation: None

Assembler

Syntax: FNOP

Attributes: Unsized

Description: This instruction does not perform any explicit operation. However, it is useful to force synchronization of the floating-point unit with an integer unit or to force processing of pending exceptions. For most floating-point instructions, the integer unit is allowed to continue with the execution of the next instruction once the floating-point unit has any operands needed for an operation, thus supporting concurrent execution of floating-point and integer instructions. The FNOP instruction synchronizes the floating-point unit and the integer unit by causing the integer unit to wait until all previous floating-point instructions have completed. Execution of FNOP also forces any exceptions pending from the execution of a previous floating-point instruction to be processed as a preinstruction exception.

The MC68882 may not wait to begin execution of another floating-point instruction until it has completed execution of the current instruction. The FNOP instruction synchronizes the coprocessor and microprocessor unit by causing the microprocessor unit to wait until the current instruction (or both instructions) have completed.

The FNOP instruction also forces the processing of exceptions pending from the execution of previous instructions. This is also inherent in the way that the floating-point coprocessor utilizes the M68000 family coprocessor interface. Once the floating-point coprocessor has received the input operand for an arithmetic instruction, it always releases the main processor to execute the next instruction (regardless of whether or not concurrent execution is prevented for the instruction due to tracing) without reporting the exception during the execution of that instruction. Then, when the main processor attempts to initiate the execution of the next floating-point coprocessor instruction, a preinstruction exception may be reported to initiate exception processing for an exception that occurred during a previous instruction. By using the FNOP instruction, the user can force any pending exceptions to be processed without performing any other operations.

Floating-Point Status Register: Not Affected.

FNOP

No Operation
 (MC6888X, MC68040)

FNOP

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

NOTE

FNOP uses the same opcode as the FBcc.W < label > instruction, with cc = F (nontrapping false) and < label > = + 2 (which results in a displacement of 0).

FREM

IEEE Remainder (MC6888X, M68040FPSP)

FREM

- Operation:** IEEE Remainder of (FPn ÷ Source) → FPn
- Assembler Syntax:** FREM. < fmt > < ea > ,FPn
FREM.X FPm,FPn
- Attributes:** Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates the modulo remainder of the number in the destination floating-point data register, using the source operand as the modulus. Stores the result in the destination floating-point data register and stores the sign and seven least significant bits of the quotient in the floating-point status register quotient byte (the quotient is the result of FPn ÷ Source). The IEEE remainder function is defined as:

$$FPn - (Source \times N)$$

where N = INT (FPn ÷ Source) in the round-to-nearest mode.

The FREM function is not defined for a source operand equal to zero or for a destination operand equal to infinity. Note that this function is not the same as the FMOD instruction, which uses the round-to-zero mode and thus returns a remainder that is different from the remainder required by the IEEE *Specification for Binary Floating-Point Arithmetic*.

Operation Table:

DESTINATION		SOURCE ¹								
		+	In Range	–	+	Zero#	–	+	Infinity	–
In Range	+	IEEE Remainder		NaN ²		FPn ²				
	–									
Zero	+	+ 0.0 – 0.0		NaN ²		+ 0.0 – 0.0				
	–									
Infinity	+	NaN ²		NaN ²		NaN† ²				
	–									

- NOTES:
1. If the source operand is a NaN, refer to **1.6.5 Not-A-Numbers** for more information.
 2. Sets the OPERR bit in the floating-point status register exception byte.
 3. Returns the value of FPn before the operation. However, the result is processed by the normal instruction termination procedure to round it as required. Thus, an overflow and/or inexact result may occur if the rounding precision has been changed to a smaller size since the FPn value was loaded.

FREM

IEEE Remainder (MC6888X, M68040FPSP)

FREM

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Loaded with the sign and least significant seven bits of the quotient ($FPn \div \text{Source}$). The sign of the quotient is the exclusive-OR of the sign bits of the source and destination operands.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the source is zero or the destination is infinity; cleared otherwise.
	OVFL	Cleared
	UNFL	Refer to underflow in the appropriate user's manual.
	DZ	Cleared
	INEX2	Cleared
	INEX1	If < fmt > is packed, refer to inexact result on decimal input in the appropriate user's manual; cleared otherwise.
Accrued Exception Byte:	Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	1	0	0	1	0	1

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

FREM

IEEE Remainder (MC6888X, M68040FPSP)

FREM

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register.

FSCALE

Scale Exponent
(MC6888X, M68040FPSP)

FSCALE

- Operation:** $FPn \times INT(2Source) \rightarrow FPn$
- Assembler Syntax:** `FSCALE. < fmt > < ea > ,FPn`
`FSCALE.X FPm,FPn`
- Attributes:** Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to an integer (if necessary) and adds that integer to the destination exponent. Stores the result in the destination floating-point data register. This function has the effect of multiplying the destination by 2^{Source} , but is much faster than a multiply operation when the source is an integer value.

The floating-point coprocessor assumes that the scale factor is an integer value before the operation is executed. If not, the value is chopped (i.e., rounded using the round-to-zero mode) to an integer before it is added to the exponent. When the absolute value of the source operand is $\geq 2^{14}$, an overflow or underflow always results.

Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	–	+	Zero	–	+	Infinity	–
In Range + –	Scale Exponent			FPn ²			NAN ³		
Zero + –	+ 0.0		– 0.0	+ 0.0		– 0.0		NAN ³	
Infinity + –	+ inf		– inf	+ inf		– inf		NAN ³	

- NOTES:
1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
 2. Returns the value of FPn before the operation. However, the result is processed by the normal instruction termination procedure to round it as required. Thus, an overflow and/or inexact result may occur if the rounding precision has been changed to a smaller size since the FPn value was loaded.
 3. Sets the OPERR bit in the floating-point status register exception byte.



FSCALE

Scale Exponent
(MC6888X, M68040FPSP)

FSCALE

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Not affected	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the source operand is $\pm$ infinity; cleared otherwise.
	OVFL	Refer to overflow in the appropriate user's manual.
	UNFL	Refer to underflow in the appropriate user's manual.
	DZ	Cleared
	INEX2	Cleared
	INEX1	If < fmt > is packed, refer to inexact result on decimal input in the appropriate user's manual; cleared otherwise.
Accrued Exception Byte:	Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	1	0	0	1	1	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

FSCALE

Scale Exponent (MC6888X, M68040FPSP)

FSCALE

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register.



FScC

Set According to Floating-Point Condition
(MC6888X, MC68040)

FScC

Operation: If (Condition True)
Then 1s → Destination
Else 0s → Destination

Assembler Syntax: FScC. < size > < ea >

Attributes: Size = (Byte)

Description: If the specified floating-point condition is true, sets the byte integer operand at the destination to TRUE (all ones); otherwise, sets the byte to FALSE (all zeros). The conditional specifier cc may select any one of the 32 floating-point conditional tests as described in **Table 3-23 Floating-Point Conditional Tests**.

Floating-Point Status Register:

Condition Codes:	Not affected.	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Set if the NAN condition code is set and the condition selected is an IEEE nonaware test.
	SNAN	Not Affected.
	OPERR	Not Affected.
	OVFL	Not Affected.
	UNFL	Not Affected.
	DZ	Not Affected.
	INEX2	Not Affected.
	INEX1	Not Affected.
Accrued Exception Byte:	The IOP bit is set if the BSUN bit is set in the exception byte. No other bit is affected.	

FScc

Set According to Floating-Point Condition (MC6888X, MC68040)

FScc

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	1	EFFECTIVE ADDRESS MODE REGISTER					
0	0	0	0	0	0	0	0	0	0	CONDITIONAL PREDICATE					

Instruction Fields:

Effective Address field—Specifies the addressing mode for the byte integer operand.
Only data alterable addressing modes can be used as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	—	—
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	—	—
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	—	—
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	—	—
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	—	—
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	—	—

Conditional Predicate field—Specifies one of 32 conditional tests as defined in **3.6.2 Conditional Testing**.

NOTE

When a BSUN exception occurs, a preinstruction exception is taken. If the exception handler returns without modifying the image of the program counter on the stack frame (to point to the instruction following the FScc), then it must clear the cause of the exception (by clearing the NAN bit or disabling the BSUN trap) or the exception occurs again immediately upon return to the routine that caused the exception.

FSGLDIV

Single-Precision Divide (MC6888X, MC68040)

FSGLDIV

Operation: $FPn \div \text{Source} \rightarrow FPn$

Assembler Syntax: `FSGLDIV.< fmt > < ea > ,FPn`
`FSGLDIV.X FPm,FPn`

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and divides that number into the number in the destination floating-point data register. Stores the result in the destination floating-point data register, rounded to single precision (despite the current rounding precision). This function is undefined for $0 \div 0$ and $\text{infinity} \div \text{infinity}$.

Both the source and destination operands are assumed to be representable in the single-precision format. If either operand requires more than 24 bits of mantissa to be accurately represented, the extraneous mantissa bits are truncated prior to the division, hence the accuracy of the result is not guaranteed. Furthermore, the result exponent may exceed the range of single precision, regardless of the rounding precision selected in the floating-point control register mode control byte. Refer to **3.6.1 Underflow, Round, Overflow** for more information.

The accuracy of the result is not affected by the number of mantissa bits required to represent each input operand since the input operands just change to extended precision. The result mantissa is rounded to single precision, and the result exponent is rounded to extended precision, despite the rounding precision selected in the floating-point control register.

Operation Table:

DESTINATION	SOURCE ^{3,1}					
		In Range	Zero	Infinity		
In Range	+	Divide (Single Precision)	$+ \text{inf}^2$	$- \text{inf}^2$	$+ 0.0$	$- 0.0$
	-		$- \text{inf}^2$	$+ \text{inf}^2$	$- 0.0$	$+ 0.0$
Zero	+	$+ 0.0$	NAN ³	$+ 0.0$	$- 0.0$	$- 0.0$
	-	$- 0.0$		$- 0.0$	$+ 0.0$	$+ 0.0$
Infinity	+	$+ \text{inf}$	$+ \text{inf}$	$- \text{inf}$	NAN ³	
	-	$- \text{inf}$	$- \text{inf}$	$+ \text{inf}$		

- NOTES:
1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
 2. Sets the DZ bit in the floating-point status register exception byte.
 3. Sets the OPERR bit in the floating-point status register exception byte.

FSGLDIV

Single-Precision Divide
(MC6888X, MC68040)

FSGLDIV

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set for $0 \div 0$ or infinity $\div$ infinity.
	OVFL	Refer to overflow in the appropriate user's manual.
	UNFL	Refer to underflow in the appropriate user's manual.
	DZ	Set if the source is zero and the destination is in range; cleared otherwise.
	INEX2	Refer to inexact result in the appropriate user's manual.
	INEX1	If $\langle \text{fmt} \rangle$ is packed, refer to the appropriate user's manual for inexact result on decimal input; cleared otherwise.
Accrued Exception Byte:	Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	1	0	0	1	0	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

FSGLDIV

Single-Precision Divide (MC6888X, MC68040)

FSGLDIV

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register.

FSGLMUL

Single-Precision Multiply
(MC6888X, MC68040)

FSGLMUL

Operation:

Source x FPN → FPN

Assembler

FSGLMUL. < fmt > < ea > ,FPn

Syntax:

FSGLMUL.X FPM,FPn

Attributes:

Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and multiplies that number by the number in the destination floating-point data register. Stores the result in the destination floating-point data register, rounded to single precision (regardless of the current rounding precision).

Both the source and destination operands are assumed to be representable in the single-precision format. If either operand requires more than 24 bits of mantissa to be accurately represented, the extraneous mantissa bits are truncated prior to the multiplication; hence, the accuracy of the result is not guaranteed. Furthermore, the result exponent may exceed the range of single precision, regardless of the rounding precision selected in the floating-point control register mode control byte. Refer to **3.6.1 Underflow, Round, Overflow** for more information.

Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	–	+	Zero	–	+	Infinity	–
In Range	+	Multiply		+	0.0	–	0.0	+	inf
	–	(Single Precision)		–	0.0		+ 0.0	–	inf
Zero	+	+	0.0	–	0.0	+	0.0	–	0.0
	–	–	0.0	+	0.0	–	0.0		+ 0.0
								NAN ²	
Infinity	+	+	inf	–	inf			+	inf
	–	–	inf	+	inf			–	inf

- NOTES:
1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
 2. Sets the OPERR bit in the floating-point status register exception byte.

NOTE

The input operand mantissas truncate to single precision before the multiply operation. The result mantissa rounds to single precision despite the rounding precision selected in the floating-point control register.

FSGLMUL

Single-Precision Multiply
(MC6888X, MC68040)

FSGLMUL

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if one operand is zero and the other is infinity; cleared otherwise.
	OVFL	Refer to overflow in the appropriate user's manual.
	UNFL	Refer to underflow in the appropriate user's manual.
	DZ	Cleared
	INEX2	Refer to inexact result in the appropriate user's manual.
	INEX1	If < fmt > is packed, refer to inexact result on decimal input in the appropriate user's manual; cleared otherwise.
Accrued Exception Byte:	Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	1	0	0	1	1	1

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

FSGLMUL

Single-Precision Multiply (MC6888X, MC68040)

FSGLMUL

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register.

FSIN

Sine (MC6888X, M68040FPSP)

FSIN

Operation: Sine of Source $\rightarrow$ FPn

Assembler FSIN. < fmt > < ea > ,FPn

Syntax: FSIN.X FPm,FPn
FSIN.X FPn

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates the sine of that number. Stores the result in the destination floating-point data register. This function is not defined for source operands of $\pm$ infinity. If the source operand is not in the range of $[-2\pi \dots +2\pi]$, the argument is reduced to within that range before the sine is calculated. However, large arguments may lose accuracy during reduction, and very large arguments (greater than approximately 10^{20}) lose all accuracy. The result is in the range of $[-1 \dots +1]$.

Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	-	+	Zero	-	+	Infinity	-
Result	Sine			+ 0.0		- 0.0		NaN ²	

NOTES:

1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
2. Sets the OPERR bit in the floating-point status register exception byte.

Floating-Point Status Register:

Condition Codes: Affected as described in **3.6.2 Conditional Testing**.

Quotient Byte: Not affected.

FSIN

Sine (MC6888X, M68040FPSP)

FSIN

Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the source is $\pm$ infinity; cleared otherwise.
	OVFL	Cleared
	UNFL	Refer to underflow in the appropriate user's manual.
	DZ	Cleared
	INEX2	Refer to inexact result in the appropriate user's manual.
	INEX1	If < fmt > is packed, refer to inexact result on decimal input in the appropriate user's manual; cleared otherwise.

Accrued Exception Byte: Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	1	1	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

FSIN

Sine (MC6888X, M68040FPSP)

FSIN

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, then the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FSINCOS

Simultaneous Sine and Cosine
(MC6888X, M68040FPSP)

FSINCOS

Operation:

Sine of Source → FPs
Cosine of Source → FPc

Assembler Syntax:

FSINCOS. < fmt > < ea > ,FPc,FPs
FSINCOS.X FPs,FPc,FPs

Attributes:

Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates both the sine and the cosine of that number. Calculates both functions simultaneously; thus, this instruction is significantly faster than performing separate FSIN and FCOS instructions. Loads the sine and cosine results into the destination floating-point data register. Sets the condition code bits according to the sine result. If FPs and FPc are specified to be the same register, the cosine result is first loaded into the register and then is overwritten with the sine result. This function is not defined for source operands of $\pm$ infinity.

If the source operand is not in the range of $[-2\pi... + 2\pi]$, the argument is reduced to within that range before the sine and cosine are calculated. However, large arguments may lose accuracy during reduction, and very large arguments (greater than approximately 10^{20}) lose all accuracy. The results are in the range of $[-1... + 1]$.

Operation Table:

DESTINATION	SOURCE ¹									
	+	In Range	–	+	Zero	–	+	Infinity	–	
FPs		Sine		+	0.0		–	0.0		NAN ²
FPc		Cosine			+	1.0				NAN ²

NOTES:

1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
2. Sets the OPERR bit in the floating-point status register exception byte.



FSINCOS

Simultaneous Sine and Cosine
(MC6888X, M68040FPSP)

FSINCOS

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing (for the sine result).	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the source is $\pm$ infinity; cleared otherwise.
	OVFL	Cleared
	UNFL	Set if a sine underflow occurs, in which case the cosine result is 1. Cosine cannot underflow. Refer to underflow in the appropriate user's manual.
	DZ	Cleared
	INEX2	Refer to inexact result in the appropriate user's manual.
	INEX1	If < fmt > is packed, refer to inexact result on decimal input in the appropriate user's manual; cleared otherwise.
Accrued Exception Byte:	Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER, FPs			0	1	1	0	DESTINATION REGISTER FPC		

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

FSINCOS

Simultaneous Sine and Cosine (MC6888X, M68040FPSP)

FSINCOS

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

Destination Register, FPc field—Specifies the destination floating- point data register, FPc. The cosine result is stored in this register.



FSINCOS

Simultaneous Sine and Cosine
(MC6888X, M68040FPSP)

FSINCOS

Destination Register, FPs field—Specifies the destination floating- point data register, FPs. The sine result is stored in this register. If FPc and FPs specify the same floating-point data register, the sine result is stored in the register, and the cosine result is discarded.

If R/M = 0 and the source register field is equal to either of the destination register fields, the input operand is taken from the specified floating-point data register, and the appropriate result is written into the same register.

FSINH

Hyperbolic Sine

(MC6888X, M68040FPSP)

FSINH

Operation:

Hyperbolic Sine of Source → FPn

Assembler

FSINH. < fmt > < ea > ,FPn

Syntax:

FSINH.X FPm,FPn

FSINH.X FPn

Attributes:

Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description:

Converts the source operand to extended precision (if necessary) and calculates the hyperbolic sine of that number. Stores the result in the destination floating-point data register.

Operation Table:

DESTINATION	SOURCE								
	+	In Range	–	+	Zero	–	+	Infinity	–
Result		Hyperbolic Sine		+	0.0	– 0.0	+	inf	– inf

NOTE: If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.

Floating-Point Status Register:

Condition Codes:

Affected as described in **3.6.2 Conditional Testing**.

Quotient Byte:

Not affected.

Exception Byte:

BSUN

Cleared

SNAN

Refer to **1.6.5 Not-A-Numbers**.

OPERR

Cleared

OVFL

Refer to overflow in the appropriate user’s manual.

UNFL

Refer to underflow in the appropriate user’s manual.

DZ

Cleared

INEX2

Refer to inexact result in the appropriate user’s manual.

INEX1

If < fmt > is packed, refer to inexact result on decimal input in the appropriate user’s manual; cleared otherwise.

Accrued Exception Byte:

Affected as described in IEEE exception and trap compatibility in the appropriate user’s manual.



FSINH

Hyperbolic Sine (MC6888X, M68040FPSP)

FSINH

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0		SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	0	0	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

Effective Address field—Determines the addressing mode for external operands.
 If R/M = 0, this field is unused and should be all zeros.
 If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

FSINH

Hyperbolic Sine (MC6888X, M68040FPSP)

FSINH

R/M field—Specifies the source operand address mode.

- 0 — The operation is register to register.
- 1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FSQRT

Floating-Point Square Root (MC6888X, MC68040)

FSQRT

Operation: Square Root of Source → FPn

Assembler Syntax: FSQRT. < fmt > < ea > ,FPn
FSQRT.X FPm,FPn
FSQRT.X FPn
*FrSQRT. < fmt > < ea > ,FPn
*FrSQRT FPm,FPn
*FrSQRT FPn
where r is rounding precision, S or D
*Supported by MC68040 only

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates the square root of that number. Stores the result in the destination floating-point data register. This function is not defined for negative operands.

FSQRT will round the result to the precision selected in the floating-point control register. FSFSQRT and FDFSQRT will round the result to single or double precision, respectively, regardless of the rounding precision selected in the floating-point control register. Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	–	+	Zero	–	+	Infinity	–
Result	$\sqrt{x}$		NAN ²	+	0.0	– 0.0	+	inf	NAN ²

- NOTES:
1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
 2. Sets the OPERR bit in the floating-point status register exception byte.

FSQRT

Floating-Point Square Root

(MC6888X, MC68040)

FSQRT

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the source operand is not zero and is negative; cleared otherwise.
	OVFL	Cleared
	UNFL	Cleared
	DZ	Cleared
	INEX2	Refer to exception processing in the appropriate user's manual.
	INEX1	If < fmt > is packed, refer to exception processing in the appropriate user's manual; cleared otherwise.
Accrued Exception Byte:	Affected as described in exception processing in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS MODE REGISTER					
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

FSQRT

Floating-Point Square Root (MC6888X, MC68040)

FSQRT

Instruction Fields:

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, specifies the location of the source operand. Only data addressing modes can be used as listed in the following table:

Addressing Mode	Mode	Register		Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn		(xxx).W	111	000
An	—	—		(xxx).L	111	001
(An)	010	reg. number:An		# < data >	111	100
(An) +	011	reg. number:An				
– (An)	100	reg. number:An				
(d ₁₆ ,An)	101	reg. number:An		(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An		(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An		(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An		([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An		([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)*

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

*This encoding will cause an unimplemented data type exception in the MC68040 to allow emulation in software.

FSQRT

Floating-Point Square Root
(MC6888X, MC68040)

FSQRT

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

Opmode field—Specifies the instruction and rounding precision.

- | | | |
|---------|--------|--|
| 0000100 | FSQRT | Rounding precision specified by the floating-point control register. |
| 1000001 | FSSQRT | Single-precision rounding specified. |
| 1000101 | FDSQRT | Double-precision rounding specified. |

FSUB

Floating-Point Subtract (MC6888X, MC68040)

FSUB

Operation: FPn – Source → FPn

Assembler

Syntax: FSUB. < fmt > < ea > ,FPn
 FSUB.X FPr,FPn
 *FrSUB. < fmt > < ea > ,FPn
 *FrSUB.X FPr,FPn
 where r is rounding precision, S or D
 *Supported by MC68040 only

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and subtracts that number from the number in the destination floating-point data register. Stores the result in the destination floating-point data register.

Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	–	+	Zero	–	+	Infinity	–
In Range	+	Subtract			Subtract		– inf		+ inf
	–								
Zero	+	Subtract			+ 0.0 ² + 0.0		– inf		+ inf
	–								
Infinity	+	+ inf – inf			+ inf – inf		NAN ² – inf		– inf NAN ²
	–								

- NOTES:
1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
 2. Returns + 0.0 in rounding modes RN, RZ, and RP; returns – 0.0 in RM.
 3. Sets the OPERR bit in the floating-point status register exception byte.

FSUB

Floating-Point Subtract (MC6888X, MC68040)

FSUB

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if both the source and destination are like-signed infinities; cleared otherwise.
	OVFL	Refer to exception processing in the appropriate user's manual.
	UNFL	Refer to exception processing in the appropriate user's manual.
	DZ	Cleared
	INEX2	Refer to exception processing in the appropriate user's manual.
	INEX1	If < fmt > is packed, refer to exception processing in the appropriate user's manual; cleared otherwise.
Accrued Exception Byte:	Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
											MODE		REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			OPMODE						

FSUB

Floating-Point Subtract (MC6888X, MC68040)

FSUB

Instruction Fields:

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, specifies the location of the source operand. Only data addressing modes can be used as listed in the following table:

Addressing Mode	Mode	Register		Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn		(xxx).W	111	000
An	—	—		(xxx).L	111	001
(An)	010	reg. number:An		# < data >	111	100
(An) +	011	reg. number:An				
– (An)	100	reg. number:An				
(d ₁₆ ,An)	101	reg. number:An		(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An		(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An		(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An		([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An		([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)*

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

*This encoding will cause an unimplemented data type exception in the MC68040 to allow emulation in software.

FSUB

Floating-Point Subtract (MC6888X, MC68040)

FSUB

Destination Register field—Specifies the destination floating- point data register.

Opmode field—Specifies the instruction and rounding precision.

0101000	FSUB	Rounding precision specified by the floating- point control register.
1101000	FSSUB	Single-precision rounding specified.
1101100	FDSUB	Double-precision rounding specified.

FTAN

Tangent (MC6888X/004SW)

FTAN

Operation: Tangent of Source $\rightarrow$ FPn

Assembler FTAN. < fmt > < ea > ,FPn

Syntax: FTAN.X FPM,FPn
FTAN.X FPn

Attributes: Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates the tangent of that number. Stores the result in the destination floating-point data register. This function is not defined for source operands of $\pm$ infinity. If the source operand is not in the range of $[-\pi/2 \dots +\pi/2]$, the argument is reduced to within that range before the tangent is calculated. However, large arguments may lose accuracy during reduction, and very large arguments (greater than approximately 10^{20}) lose all accuracy.

Operation Table:

DESTINATION	SOURCE ¹								
	+	In Range	-	+	Zero	-	+	Infinity	-
Result	Tangent			+ 0.0		- 0.0	NAN ²		

NOTES:

1. If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.
2. Sets the OPERR bit in the floating-point status register exception byte.

FTAN

Tangent
(MC6888X/004SW)

FTAN

Floating-Point Status Register:

Condition Codes:	Affected as described in 3.6.2 Conditional Testing .	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Cleared
	SNAN	Refer to 1.6.5 Not-A-Numbers .
	OPERR	Set if the source is $\pm$ infinity; cleared otherwise.
	OVFL	Refer to overflow in the appropriate user's manual.
	UNFL	Refer to underflow in the appropriate user's manual.
	DZ	Cleared
	INEX2	Refer to inexact result in the appropriate user's manual.
	INEX1	If < fmt > is packed, refer to inexact result on decimal input in the appropriate user's manual; cleared otherwise.
Accrued Exception Byte:	Affected as described in IEEE exception and trap compatibility in the appropriate user's manual.	

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	1	1	1

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

FTAN

Tangent (MC6888X/004SW)

FTAN

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

000 — Long-Word Integer (L)

001 — Single-Precision Real (S)

010 — Extended-Precision Real (X)

011 — Packed-Decimal Real (P)

100 — Word Integer (W)

101 — Double-Precision Real (D)

110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FTANH

Hyperbolic Tangent

(MC6888X, M68040FPSP)

FTANH

Operation:

Hyperbolic Tangent of Source → FPn

Assembler

FTANH. < fmt > < ea > ,FPn

Syntax:

FTANH.X Fm,FPn

FTANH.X FPn

Attributes:

Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description:

Converts the source operand to extended precision (if necessary) and calculates the hyperbolic tangent of that number. Stores the result in the destination floating-point data register.

Operation Table:

DESTINATION	SOURCE								
	+	In Range	–	+	Zero	–	+	Infinity	–
Result		Hyperbolic Tangent		+	0.0	– 0.0	+	1.0	– 1.0

NOTE: If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.

Floating-Point Status Register:

Condition Codes:

Affected as described in **3.6.2 Conditional Testing**.

Quotient Byte:

Not affected.

Exception Byte:

BSUN

Cleared

SNAN

Refer to **1.6.5 Not-A-Numbers**.

OPERR

Cleared

OVFL

Cleared

UNFL

Refer to underflow in the appropriate user’s manual.

DZ

Cleared

INEX2

Refer to inexact result in the appropriate user’s manual.

INEX1

If < fmt > is packed, refer to inexact result on decimal input in the appropriate user’s manual; cleared otherwise.

Accrued Exception Byte:

Affected as described in IEEE exception and trap compatibility in the appropriate user’s manual.



FTANH

Hyperbolic Tangent (MC6888X, M68040FPSP)

FTANH

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	0	1	0	0	1

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn
An	—	—
(An)	010	reg. number:An
(An) +	011	reg. number:An
– (An)	100	reg. number:An
(d ₁₆ ,An)	101	reg. number:An
(d ₈ ,An,Xn)	110	reg. number:An
(bd,An,Xn)	110	reg. number:An
([bd,An,Xn],od)	110	reg. number:An
([bd,An],Xn,od)	110	reg. number:An

Addressing Mode	Mode	Register
(xxx).W	111	000
(xxx).L	111	001
# < data >	111	100
(d ₁₆ ,PC)	111	010
(d ₈ ,PC,Xn)	111	011
(bd,PC,Xn)	111	011
([bd,PC,Xn],od)	111	011
([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

R/M field—Specifies the source operand address mode.

0 — The operation is register to register.

1 — The operation is < ea > to register.

FTANH

Hyperbolic Tangent (MC6888X, M68040FPSP)

FTANH

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating-point data register. If R/M = 0 and the source and destination fields are equal, the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.



FTENTOX

10^x
(MC6888X, M68040FPSP)

FTENTOX

- Operation:** 10^{Source} → FPn
- Assembler Syntax:** FTENTOX. < fmt > < ea > ,FPn
FTENTOX.X Fp_m,FP_n
FTENTOX.X FP_n
- Attributes:** Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates 10 to the power of that number. Stores the result in the destination floating-point data register.

Operation Table:

DESTINATION	SOURCE								
	+	In Range	–	+	Zero	–	+	Infinity	–
Result		10 ^x			+ 1.0			+ inf	+ 0.0

NOTE: If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.

Floating-Point Status Register:

- Condition Codes: Affected as described in **3.6.2 Conditional Testing**.
- Quotient Byte: Not affected.
- Exception Byte:
- BSUN Cleared
 - SNAN Refer to **1.6.5 Not-A-Numbers**.
 - OPERR Cleared
 - OVFL Refer to overflow in the appropriate user’s manual.
 - UNFL Refer to underflow in the appropriate user’s manual.
 - DZ Cleared
 - INEX2 Refer to inexact result in the appropriate user’s manual.
 - INEX1 If < fmt > is packed, refer to the appropriate user’s manual inexact result on decimal input; cleared otherwise.
- Accrued Exception Byte: Affected as described in IEEE exception and trap compatibility in the appropriate user’s manual.

FTENTOX

10^x

(MC6888X, M68040FPSP)

FTENTOX

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	0	0	1	0

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

Effective Address field—Determines the addressing mode for external operands.
 If R/M = 0, this field is unused and should be all zeros.
 If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

FTENTOX

10^x
(MC6888X, M68040FPSP)

FTENTOX

R/M field—Specifies the source operand address mode.

- 0 — The operation is register to register.
- 1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.

FTRAPcc

Trap on Floating-Point Condition
(MC6888X, MC68040)

FTRAPcc

Operation: If Condition True
 Then TRAP

Assembler FTRAPcc
Syntax: FTRAPcc.W # < data >
 FTRAPcc.L # < data >

Attributes: Size = (Word, Long)

Description: If the selected condition is true, the processor initiates exception processing. A vector number is generated to reference the TRAPcc exception vector. The stacked program counter points to the next instruction. If the selected condition is not true, there is no operation performed and execution continues with the next instruction in sequence. The immediate data operand is placed in the word(s) following the conditional predicate word and is available for user definition for use within the trap handler.

The conditional specifier cc selects one of the 32 conditional tests defined in **3.6.2 Conditional Testing**.

Floating-Point Status Register:

Condition Codes:	Not affected.	
Quotient Byte:	Not affected.	
Exception Byte:	BSUN	Set if the NAN condition code is set and the condition selected is an IEEE nonaware test.
	SNAN	Not Affected.
	OPERR	Not Affected.
	OVFL	Not Affected.
	UNFL	Not Affected.
	DZ	Not Affected.
	INEX2	Not Affected.
	INEX1	Not Affected.
Accrued Exception Byte:	The IOP bit is set if the BSUN bit is set in the exception byte; no other bit is affected.	



FTRAPcc

Trap on Floating-Point Condition
(MC6888X, MC68040)

FTRAPcc

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	1	1	1	1	MODE		
0	0	0	0	0	0	0	0	0	0	CONDITIONAL PREDICATE					
16-BIT OPERAND OR MOST SIGNIFICANT WORD OF 32-BIT OPERAND (IF NEEDED)															
LEAST SIGNIFICANT WORD OR 32-BIT OPERAND (IF NEEDED)															

Instruction Fields:

- Mode field—Specifies the form of the instruction.

010 — The instruction is followed by a word operand.
011 — The instruction is followed by a long-word operand.
100 — The instruction has no operand.
- Conditional Predicate field—Specifies one of 32 conditional tests as described in **3.6.2 Conditional Testing**.
- Operand field—Contains an optional word or long-word operand that is user defined.

NOTE

When a BSUN exception occurs, a preinstruction exception is taken by the main processor. If the exception handler returns without modifying the image of the program counter on the stack frame (to point to the instruction following the FTRAPcc), it must clear the cause of the exception (by clearing the NAN bit or disabling the BSUN trap), or the exception occurs again immediately upon return to the routine that caused the exception.

FTST

Test Floating-Point Operand
(MC6888X, MC68040)

FTST

Operation:

Condition Codes for Operand → FPCC

Assembler Syntax:

FTST. < fmt > < ea >
FTST.X FpM

Attributes:

Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description:

Converts the source operand to extended precision (if necessary) and sets the condition code bits according to the data type of the result.

Operation Table: The contents of this table differ from the other operation tables. A letter in an entry of this table indicates that the designated condition code bit is always set by the FTST operation. All unspecified condition code bits are cleared during the operation.

DESTINATION	SOURCE								
	+	In Range	–	+	Zero	–	+	Infinity	–
Result	none		N	Z		NZ	I		NI

NOTE: If the source operand is a NAN, set the NAN condition code bit. If the source operand is an SNAN, set the SNAN bit in the floating-point status register exception byte

Floating-Point Status Register:

Condition Codes:

Affected as described in **3.6.2 Conditional Testing**.

Quotient Byte:

Not affected.

Exception Byte:

BSUN

Cleared

SNAN

Refer to **1.6.5 Not-A-Numbers**.

OPERR

Cleared

OVFL

Cleared

UNFL

Cleared

DZ

Cleared

INEX2

Cleared

INEX1

If < fmt > is packed, refer to exception processing in the appropriate user’s manual; cleared otherwise.

Accrued Exception Byte:

Affected as described in exception processing in the appropriate user’s manual.



FTST

Test Floating-Point Operand (MC6888X, MC68040)

FTST

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	1	1	1	0	1	0

Instruction Fields:

Effective Address field—Determines the addressing mode for external operands.
 If R/M = 0, this field is unused and should be all zeros.
 If R/M = 1, specifies the location of the source operand. Only data addressing modes can be used as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.

FTST

Test Floating-Point Operand (MC6888X, MC68040)

FTST

R/M field—Specifies the source operand address mode.

- 0 — The operation is register to register.
- 1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)*
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

*This encoding will cause an unimplemented data type exception in the MC68040 to allow emulation in software.

Destination Register field—Since the floating-point unit uses a common command word format for all of the arithmetic instructions (including FTST), this field is treated in the same manner for FTST as for the other arithmetic instructions, even though the destination register is not modified. This field should be set to zero to maintain compatibility with future devices; however, the floating-point unit does not signal an illegal instruction trap if it is not zero.

FTWOTOX

2^x
(MC6888X, M68040FPSP)

FTWOTOX

- Operation:** $2^{\text{Source}} \rightarrow \text{FPn}$
- Assembler Syntax:** FTWOTOX. < fmt > < ea > ,FPn
FTWOTOX.X FPM,FPn
FTWOTOX.X FPN
- Attributes:** Format = (Byte, Word, Long, Single, Double, Extended, Packed)

Description: Converts the source operand to extended precision (if necessary) and calculates two to the power of that number. Stores the result in the destination floating-point data register.

Operation Table:

DESTINATION	SOURCE								
	+	In Range	-	+	Zero	-	+	Infinity	-
Result		2 ^x			+ 1.0		+ inf		+ 0.0

NOTE: If the source operand is a NAN, refer to **1.6.5 Not-A-Numbers** for more information.

Floating-Point Status Register:

- Condition Codes: Affected as described in **3.6.2 Conditional Testing**.
- Quotient Byte: Not affected.
- Exception Byte:
- BSUN Cleared
 - SNAN Refer to **1.6.5 Not-A-Numbers**.
 - OPERR Cleared
 - OVFL Refer to overflow in the appropriate user’s manual.
 - UNFL Refer to underflow in the appropriate user’s manual.
 - DZ Cleared
 - INEX2 Refer to inexact result in the appropriate user’s manual.
 - INEX1 If < fmt > is packed, refer to inexact result on decimal input in the appropriate user’s manual; cleared otherwise.
- Accrued Exception Byte: Affected as described in IEEE exception and trap compatibility in the appropriate user’s manual.

FTWOTOX

2^x
 (MC6888X, M68040FPSP)

FTWOTOX

Instruction Format:

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
1	1	1	1	COPROCESSOR ID			0	0	0	EFFECTIVE ADDRESS					
										MODE			REGISTER		
0	R/M	0	SOURCE SPECIFIER			DESTINATION REGISTER			0	0	1	0	0	0	1

Instruction Fields:

Coprocessor ID field—Specifies which coprocessor in the system is to execute this instruction. Motorola assemblers default to ID = 1 for the floating-point coprocessor.

Effective Address field—Determines the addressing mode for external operands.

If R/M = 0, this field is unused and should be all zeros.

If R/M = 1, this field is encoded with an M68000 family addressing mode as listed in the following table:

Addressing Mode	Mode	Register	Addressing Mode	Mode	Register
Dn*	000	reg. number:Dn	(xxx).W	111	000
An	—	—	(xxx).L	111	001
(An)	010	reg. number:An	# < data >	111	100
(An) +	011	reg. number:An			
– (An)	100	reg. number:An			
(d ₁₆ ,An)	101	reg. number:An	(d ₁₆ ,PC)	111	010
(d ₈ ,An,Xn)	110	reg. number:An	(d ₈ ,PC,Xn)	111	011
(bd,An,Xn)	110	reg. number:An	(bd,PC,Xn)	111	011
([bd,An,Xn],od)	110	reg. number:An	([bd,PC,Xn],od)	111	011
([bd,An],Xn,od)	110	reg. number:An	([bd,PC],Xn,od)	111	011

*Only if < fmt > is byte, word, long, or single.



FTWOTOX

2^x
(MC6888X, M68040FPSP)

FTWOTOX

R/M field—Specifies the source operand address mode.

- 0 — The operation is register to register.
- 1 — The operation is < ea > to register.

Source Specifier field—Specifies the source register or data format.

If R/M = 0, specifies the source floating-point data register.

If R/M = 1, specifies the source data format:

- 000 — Long-Word Integer (L)
- 001 — Single-Precision Real (S)
- 010 — Extended-Precision Real (X)
- 011 — Packed-Decimal Real (P)
- 100 — Word Integer (W)
- 101 — Double-Precision Real (D)
- 110 — Byte Integer (B)

Destination Register field—Specifies the destination floating- point data register. If R/M = 0 and the source and destination fields are equal, the input operand is taken from the specified floating-point data register, and the result is written into the same register. If the single register syntax is used, Motorola assemblers set the source and destination fields to the same value.